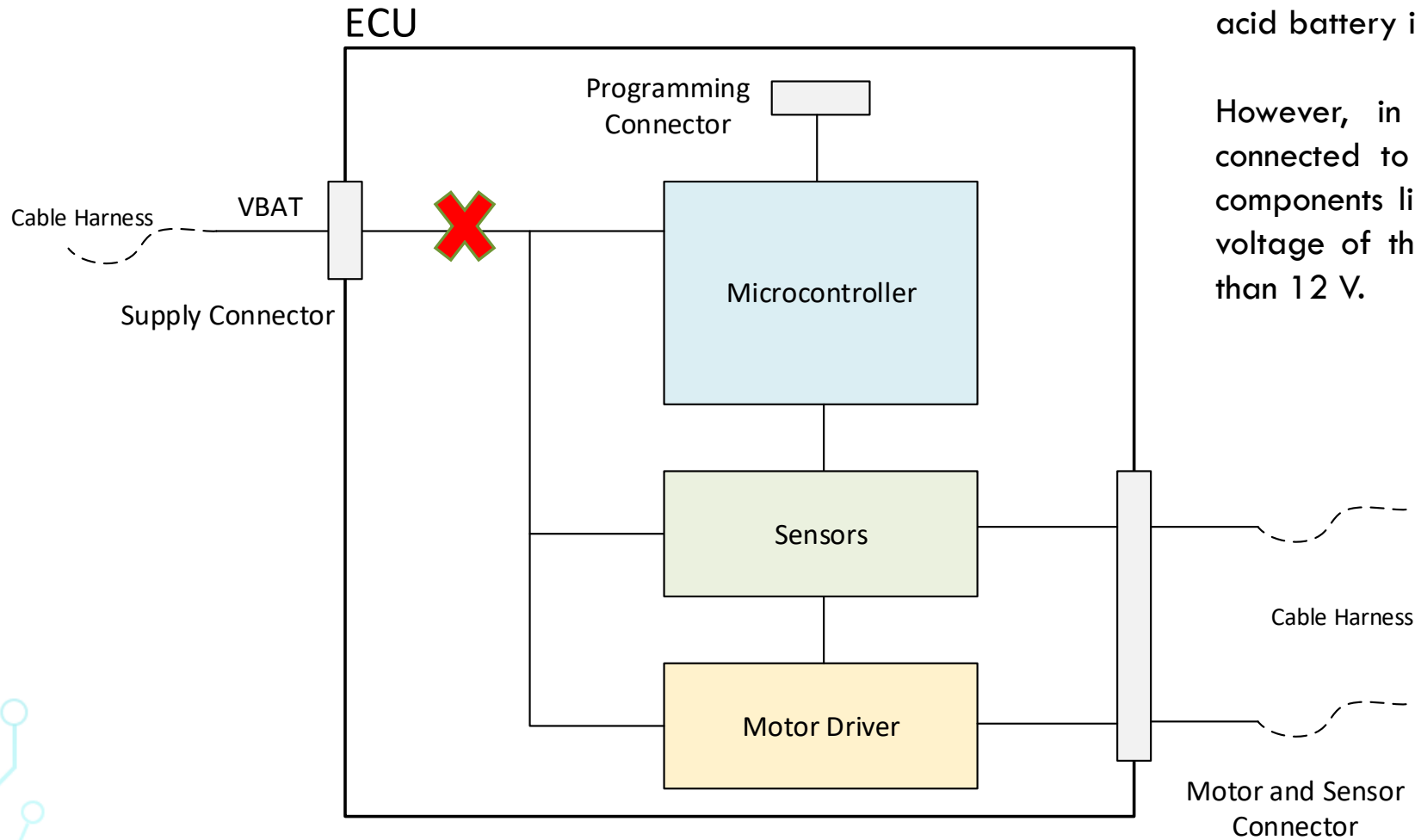


LDOs and DCDC Converters

-
- Power Supply Systems
 - LDOs (Low Dropout)
 - DCDC Converters



POWER SUPPLY SYSTEMS



The nominal battery voltage V_{BAT} of a standard lead acid battery is 12V (for heavy duty vehicles 24 V!).

However, in most cases V_{BAT} **cannot** be directly connected to the microcontrollers or other electronic components like sensors etc, since the maximum rated voltage of these electronic components is much lower than 12 V.

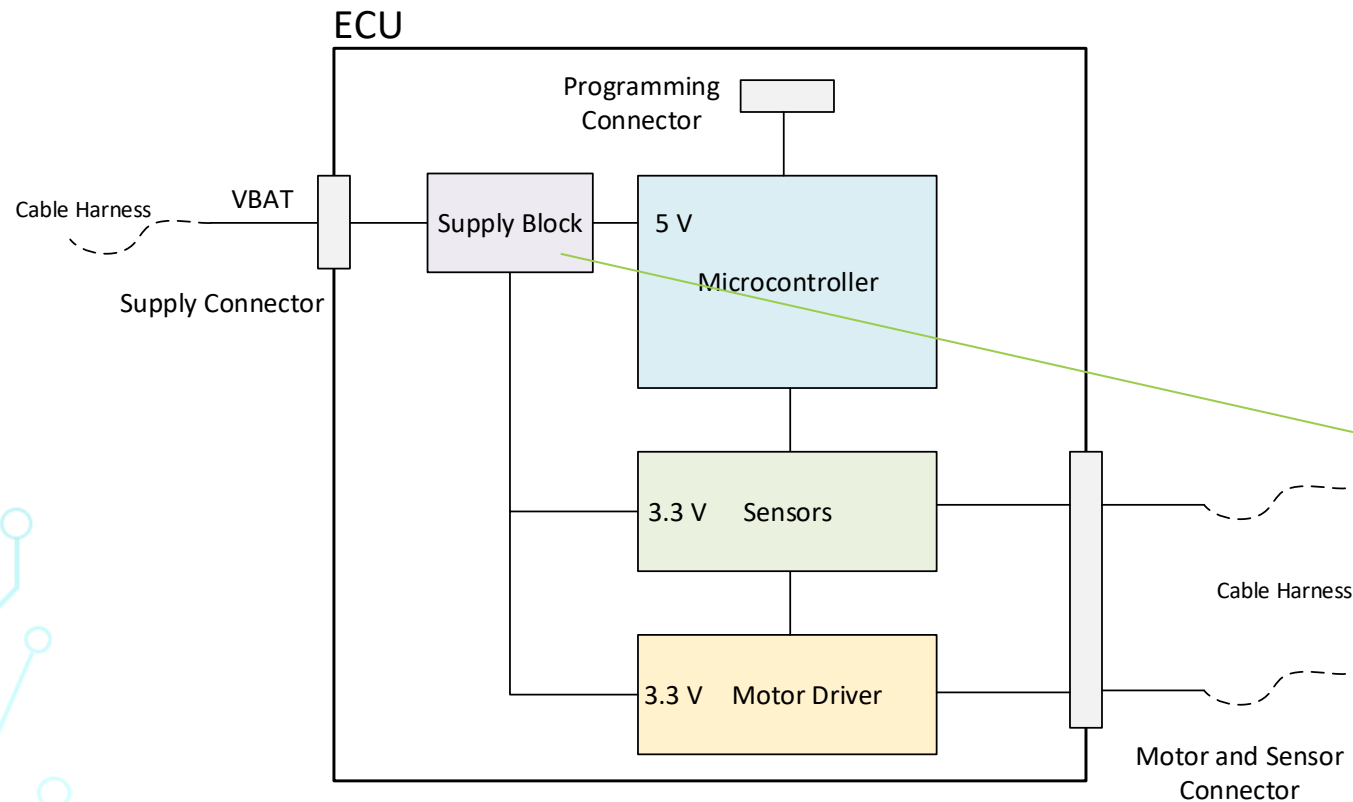
Example: Outer Mirror Control ECU (Automotive Electronic Control Unit)

POWER SUPPLY SYSTEMS

Table 37-1. Absolute Maximum Ratings

Parameter	Condition	Rating	Units
Ambient temperature under bias		-40 to +125	°C
Storage temperature		-65 to +150	°C
Voltage on pins with respect to GND			
• On the V_{DD} pin		-0.3 to +6.5	V

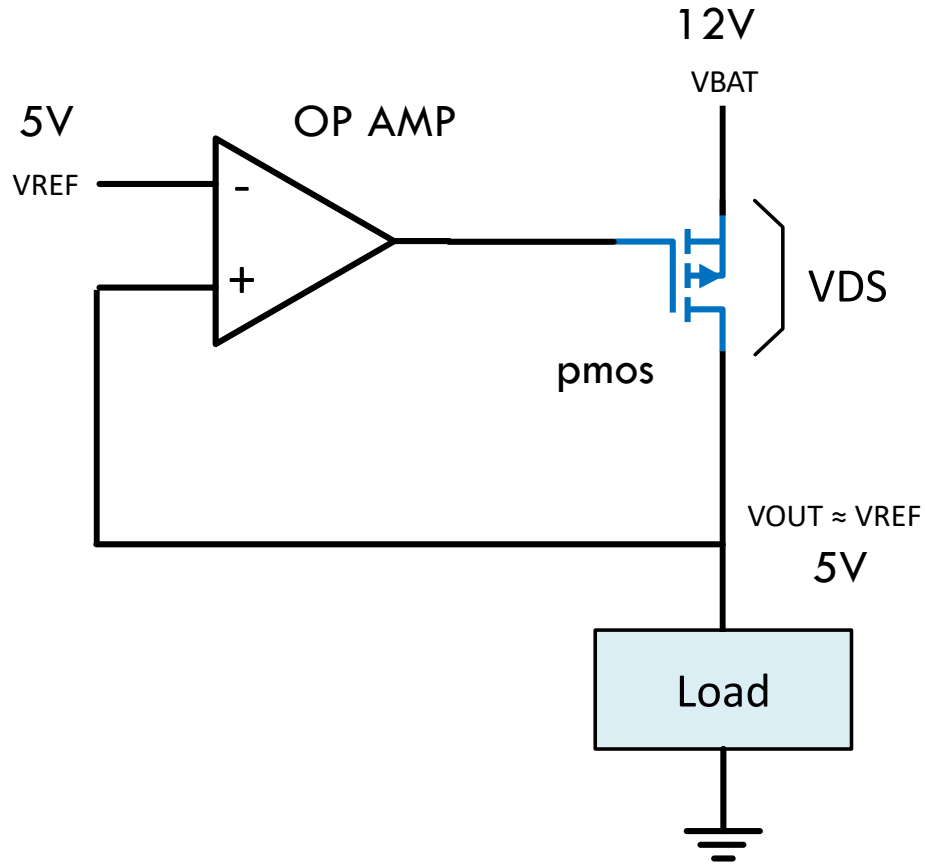
Absolute Maximum Ratings of AVR128DA28 microcontroller



The microcontroller will be damaged if the microcontroller is supplied by V_{BAT}

Additional Supply Circuits necessary to supply electronic components with different supply requirements: e.g. 5 V, 3.3 V etc

LDO



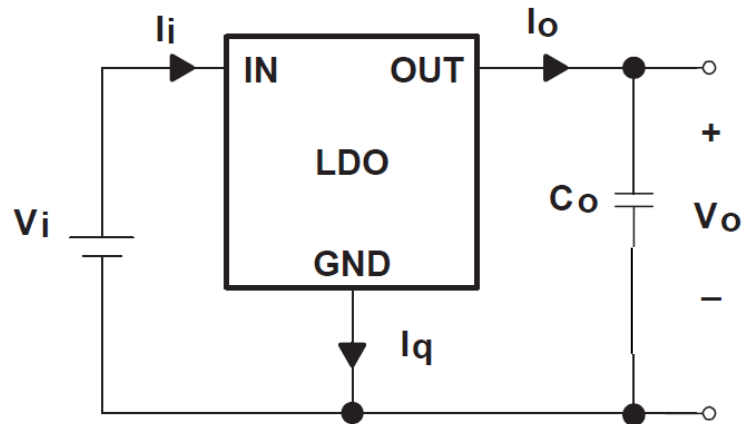
$$V_{OUT} = V_{BAT} - V_{DS} \approx V_{REF}$$

Load: microcontroller, sensor, communication interface etc.

The op amp regulates the output voltage. The residual voltage drops on MOSFET.

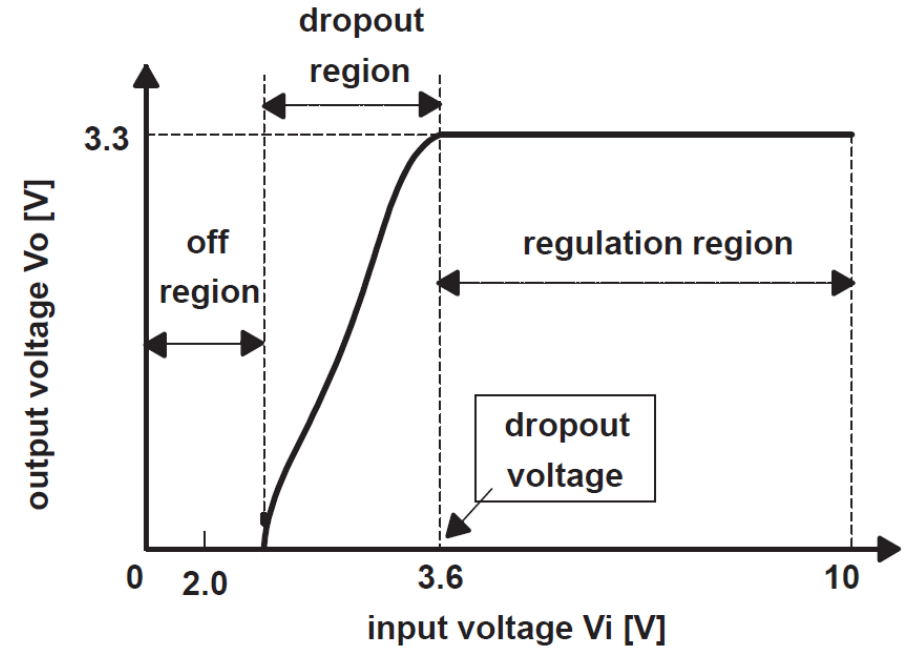
Discrete LDO: LDO designed with an opamp and a MOSFET. An ideal op amp, which does not need any supply voltage, is used.

LDO



Quiescent Current::

$$I_q = I_i - I_o$$



If bias current I_q not negligible

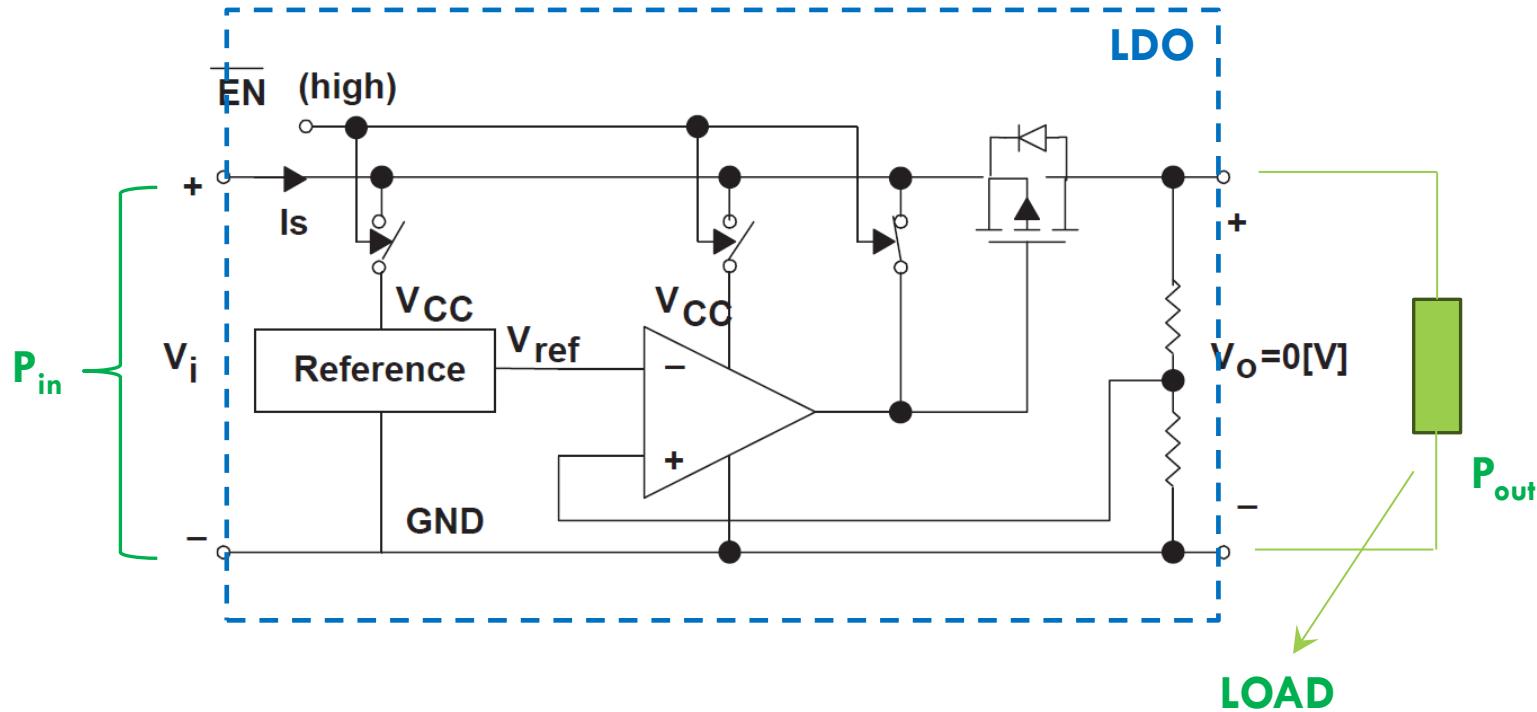


$$\text{Efficiency} = \frac{I_o V_o}{(I_o + I_q) V_i} \times 100$$

Quiescent, or ground current, is the difference between input and output currents. Low quiescent current is necessary to maximize the current efficiency.

Quiescent current consists of bias current (such as band-gap reference, sampling resistor, and error amplifier currents) and the gate drive current of the series pass element, which do not contribute to output power.

LDO

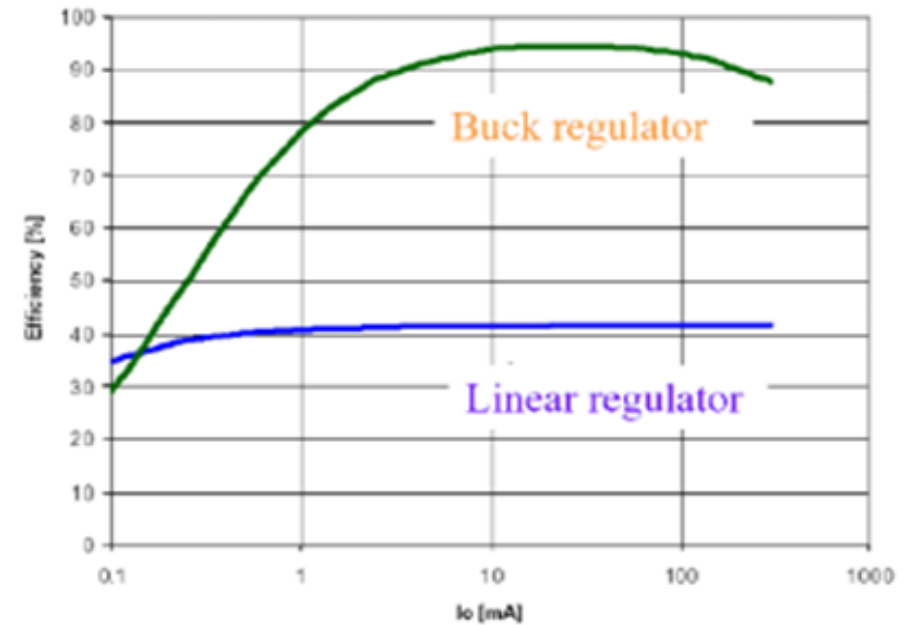


e.g. If $V_{OUT} = 5\text{ V}$, $V_{IN} = 12\text{ V}$

Efficiency: 41.67% → quite low!

58.3% of input power ($V_{IN} \cdot I_{IN}$) is dissipated at MOSFET → increase of MOSFET temperature

$$P_{\text{loss, MOSFET}} = V_{DS} \cdot I_{IN} \text{ [W]}$$



$$Eff = \frac{P_{OUT}}{P_{IN}} = \frac{V_{OUT} \cdot I_{OUT}}{V_{IN} \cdot I_{IN}}$$

$$I_{OUT} \sim I_{IN}$$

$$\Rightarrow Eff = \frac{V_{OUT}}{V_{IN}}$$

Bias currents are negligible, except for very low load current

- Power dissipation in FET: $P_{DIS} = (V_{IN} - V_{OUT}) \cdot I_{OUT}$
- The efficiency of an LDO is almost completely determined by the ratio V_{OUT} / V_{IN} .

LDO

The transient response refers to the maximum permissible variation in the output voltage caused by a step change in the load current. It depends on several parameters, including the output capacitor value (C_o), the equivalent series resistance (ESR) of the output capacitor, the bypass capacitor (C_b), which is commonly placed in parallel with the output capacitor to enhance the load transient performance, and the maximum load current ($I_{o,max}$). The maximum transient voltage deviation can be expressed as follows:

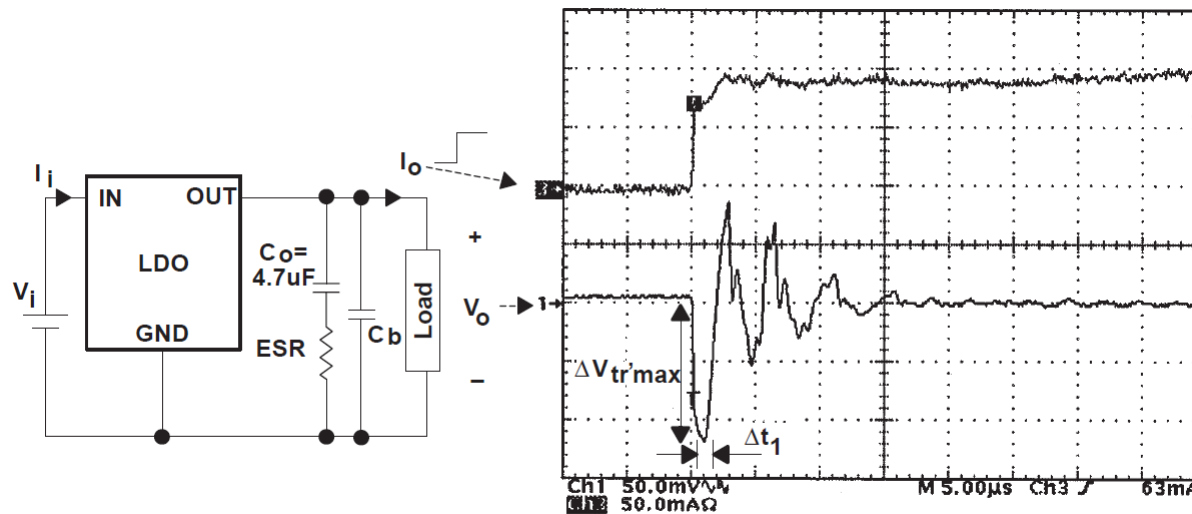
$$\Delta V_{tr,max} = \frac{I_{o,max}}{C_o + C_b} \Delta t_1 + \Delta V_{ESR}$$

$I_{o,max} (\Delta I)$ is the maximum change of the current

Δt_1 is related to loop bandwidth and represents the time it takes for the control loop to react to the load step.

$$\Delta t_1 = \frac{1}{2\pi f_{3dB}}$$

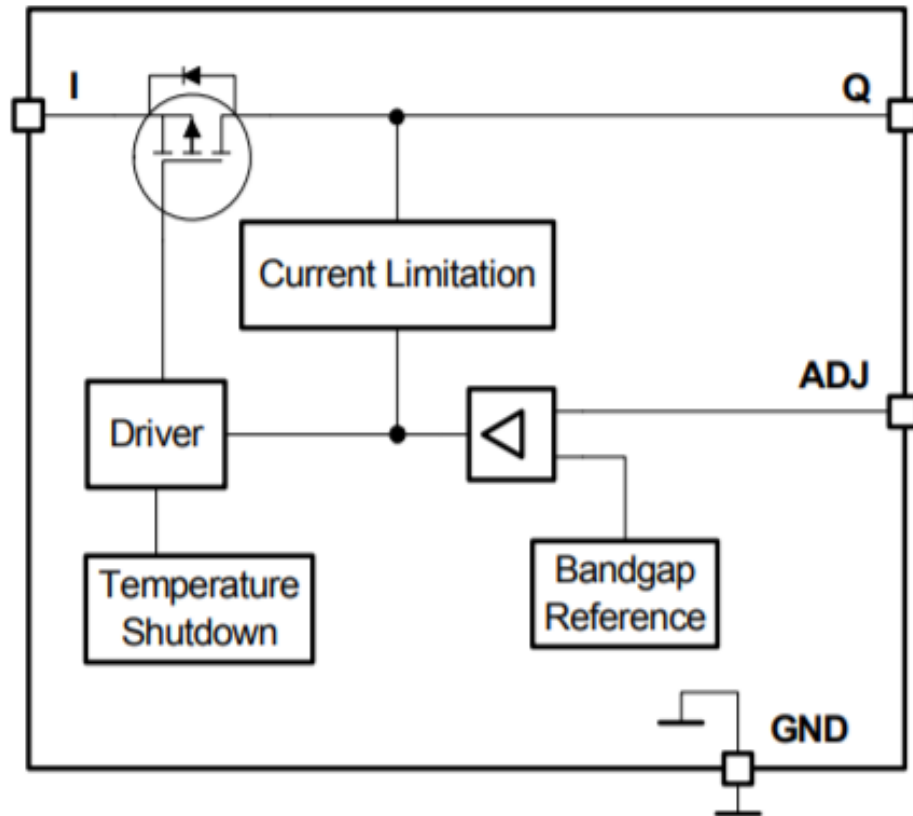
Here, Δt_1 is related to the closed-loop bandwidth f_{3dB} of the LDO regulator. ΔV_{ESR} denotes the voltage deviation caused by the equivalent series resistance (R_{ESR}) of the output capacitor. The required magnitude of this parameter depends on the specific requirements of the application.



Load Current Increased
abruptly → Step
Function

LDO

LDOs are also available in a single package. This is a more compact solution reducing the total PCB area with built-in protection mechanisms like overtemperature and overcurrent protection



Features

- Adjustable Output Voltage from 1.2 V to 5.25 V
- Output Voltage Accuracy of $\pm 3\%$
- Output Currents up to 150 mA
- Extended Input Voltage Operating Range of 2.7 V to 18 V
- Low Dropout Voltage: typ. 290mV
- Very Low Current Consumption: typ. 50 μ A
- Output Current Limitation
- Short Circuit protected
- Overtemperature Shutdown
- Wide Temperature Range From $-40\text{ }^{\circ}\text{C}$ up to $150\text{ }^{\circ}\text{C}$
- Suitable for Use in Automotive Electronics as Post Regulator
- Green Product (RoHS compliant)
- AEC Qualified

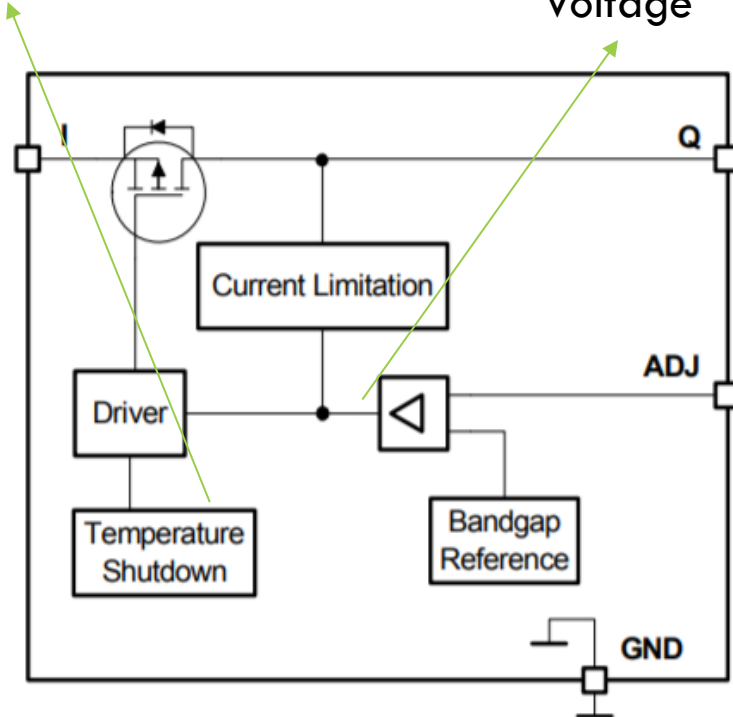
TLS202A1 from Infineon Technologies

LDO

Temperature

sensor

Reference
Voltage



TLS202A1 from Infineon Technologies

In case of overtemperature or overcurrent, the driver turns off the pass MOSFET (pmos).

Table 2 Functional Range

Parameter	Symbol	Values			Unit	Note / Test Condition	Number
		Min.	Typ.	Max.			
Input voltage	V_I	2.7	–	18	V	–	P_4.2.1
Output Capacitor Requirements for Stability	C_Q	1	–	–	μF	¹⁾	P_4.2.2
Output Capacitor Requirements for Stability	$ESR(C_Q)$	–	–	10	Ω	²⁾	P_4.2.3
Junction temperature	T_j	–40	–	150	$^{\circ}\text{C}$	–	P_4.2.4

1) The minimum output capacitance requirement is applicable for a worst case capacitance tolerance of 30%

2) relevant ESR value at $f = 10 \text{ kHz}$

Table 3 Thermal Resistance

Parameter	Symbol	Values			Unit	Note / Test Condition	Number
		Min.	Typ.	Max.			
Junction to Ambient	R_{thJA}	–	81	–	K/W	2s2p board ¹⁾	P_4.3.1
Junction to Ambient	R_{thJA}	–	217	–	K/W	Footprint only ²⁾	P_4.3.2
Junction to Ambient	R_{thJA}	–	117	–	K/W	300 mm ² PCB heatsink area ²⁾	P_4.3.3
Junction to Ambient	R_{thJA}	–	103	–	K/W	600 mm ² PCB heatsink area ²⁾	P_4.3.4
Junction to Soldering Point	R_{thJSP}	–	30	–	K/W	Pins 2, 5 fixed to T_A	P_4.3.5

1) Specified R_{thJA} value is according to JESD51-2,-5,-7 at natural convection on FR4 2s2p board; The product (chip+package) was simulated on a 76.2 x 114.3 x 1.5 mm board with 2 inner copper layers (2 x 70 μm Cu, 2 x 35 μm Cu). Where applicable a thermal via array next to the package contacted to the first inner copper layer.

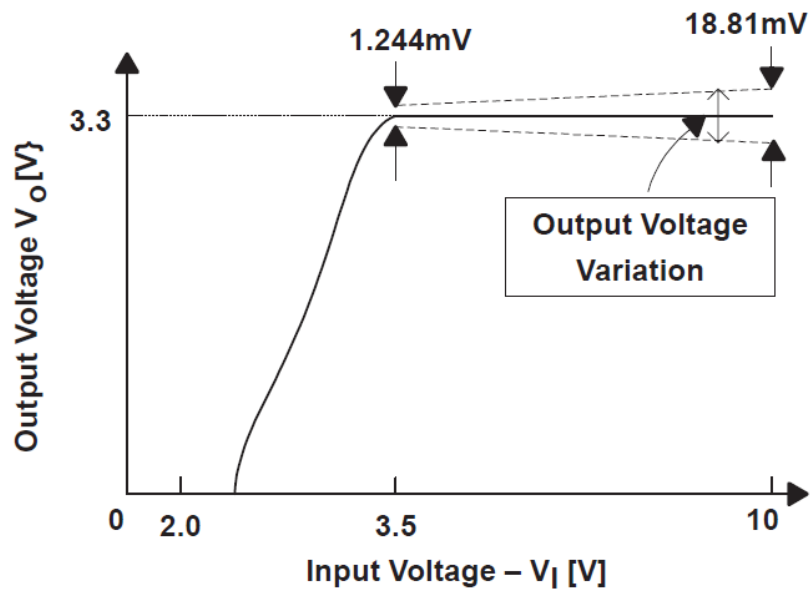
2) Package mounted on PCB FR4; 80 x 80 x 1.5 mm; 35 μm Cu, 5 μm Sn; horizontal position; zero airflow. Not subject to production test; specified by design.

2s2p: 2 signal & 2 power layers

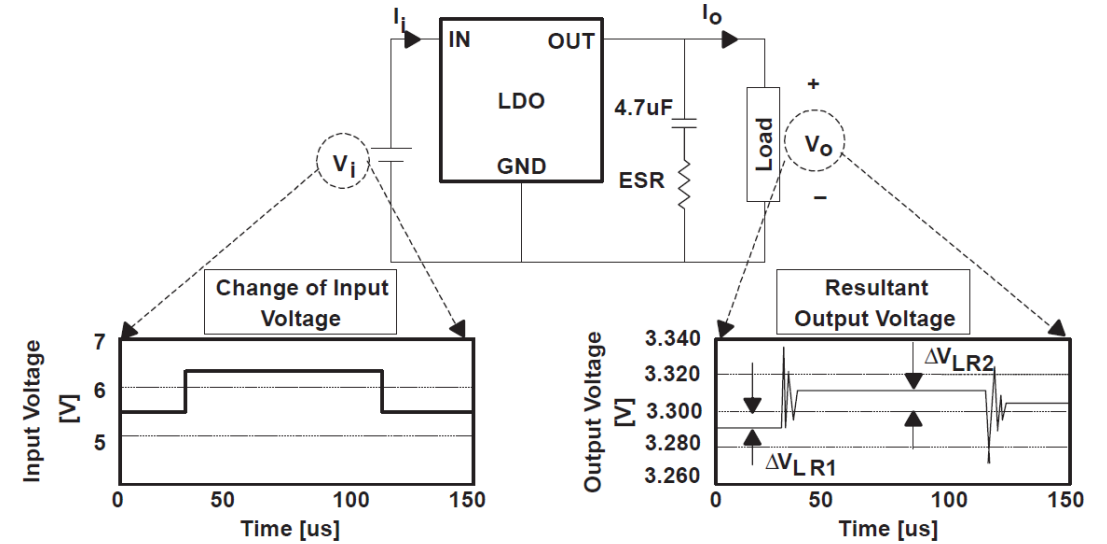
LDO

Line regulation refers to the ability of a circuit to keep the output voltage at its specified value when the input voltage changes. It is defined as:

$$\text{Line regulation} = \frac{\Delta V_o}{\Delta V_i}$$



Output Voltage With Respect to the Input Voltages



Line Transient Response

$$LR = \frac{\Delta V_{out}}{\Delta V_{in}}$$

$$\Delta V_{in} = V_{in2} - V_{in1}$$

$$\Delta V_{in} = 6.5 - 5.5$$

$$\Delta V_{in} = 1$$

$$\Delta V_{out} = V_{out2} - V_{out1}$$

$$\Delta V_{out} = 3.31 - 3.29$$

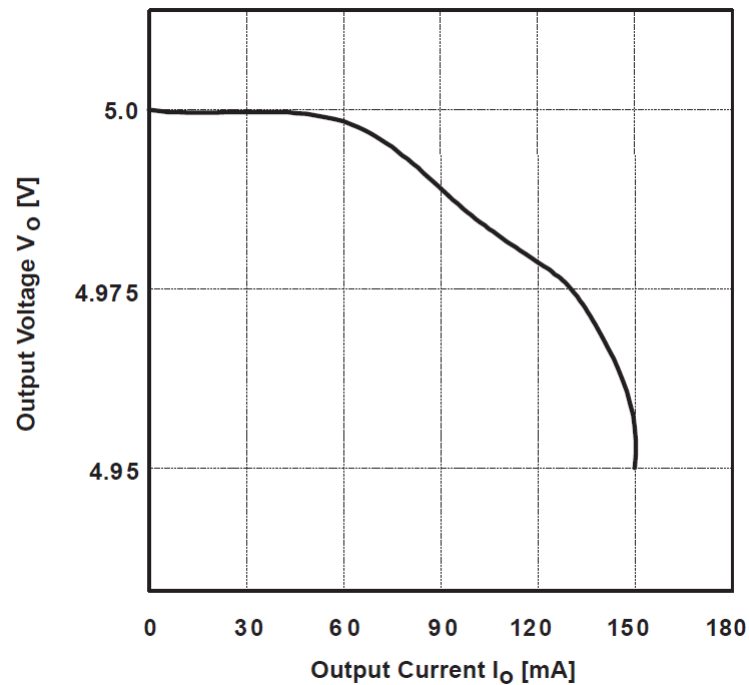
$$\Delta V_{out} = 0.02$$

$$LR = 20 \text{ mV/V}$$

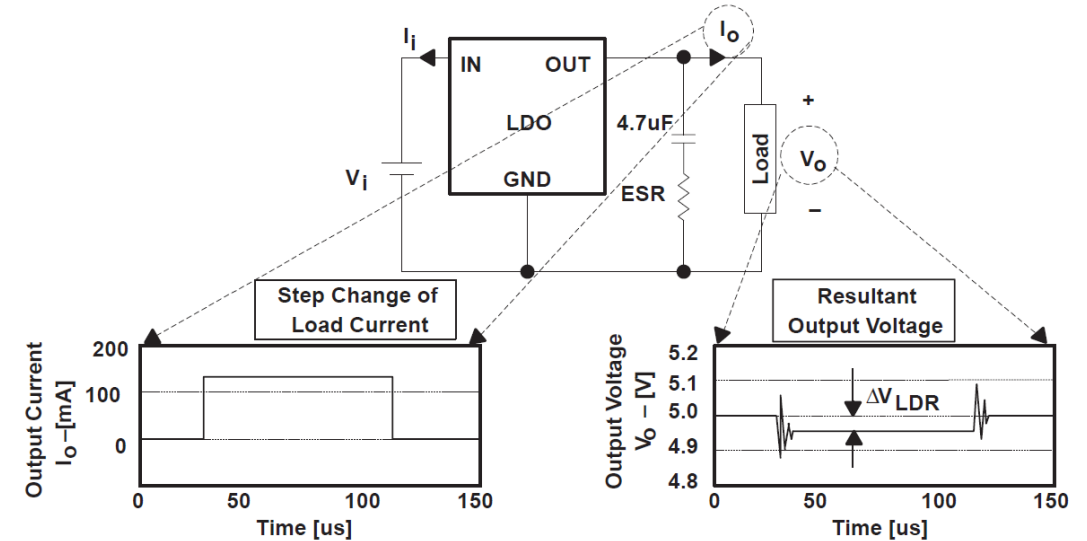
LDO

Load regulation describes the capability of a circuit to maintain the specified output voltage when the load conditions change. It is defined as:

$$\text{Load regulation; } \frac{\Delta V_o}{\Delta I_o}$$



Output Voltage With Respect to the Output Current



Load Transient Response

$$\text{LDR} = \frac{\Delta V_{out}}{\Delta I_{load}}$$



$$\begin{cases} \Delta I_{load} = I_2 - I_1 \\ \Delta I_{load} = 125 \\ \Delta V_{out} = V_{out2} - V_{out1} \\ \Delta V_{out} = 4.95 - 5.00 \\ \Delta V_{out} = -0.05 \end{cases}$$



$$\begin{aligned} |\text{LDR}| &= 0.0004 \\ \text{LDR} &= 0.4 \text{ mV/mA} \end{aligned}$$

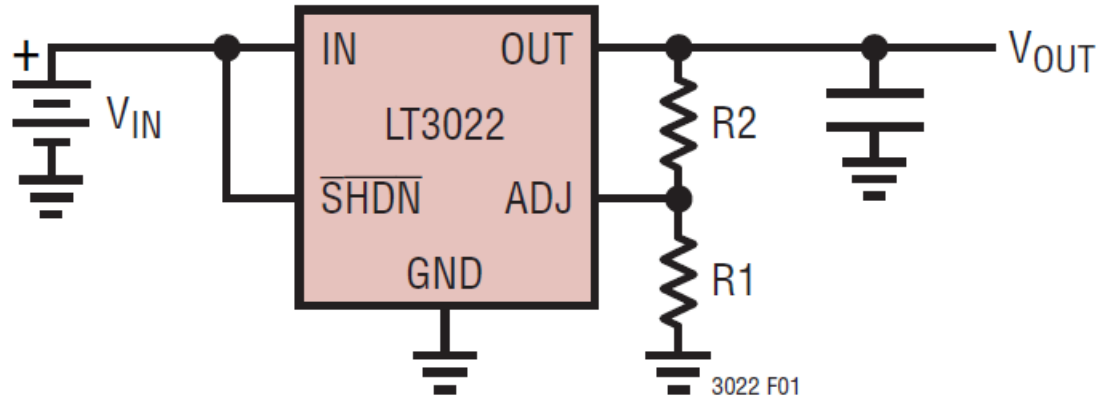
LDO – APPLICATION EXAMPLE

- Design a discrete LDO which generates 5 V from a 12 V supply. Select a MOSFET which can deliver at least 100 mA
- The LDO shall supply a microcontroller which draws 100 mA from 5 V. Model the load with a current source
- Sweep the input voltage from battery from 9 to 12 V and plot the out voltage
- The LDO may supply electronic components like microcontrollers, gate drivers or communication interfaces which have internal switching circuits (PWM generators, clock generators, CAN transceivers etc.). To model these switching components, change the current source to PWM current source with following parameters: $I_{\text{peak}} = 100 \text{ mA}$, frequency = 1 kHz, duty cycle = 50%.

Observe and try to interpret the LDO output voltage V_{OUT}

- Connect a capacitor at V_{OUT} and try to decrease the output voltage ripple to 50 mV. Why did the capacitor help? Consider using the equation: $I = C \cdot dV/dt$ while interpreting the results. Draw the current flowing through the MOSFET and the capacitor
- What limits the maximum current that can be delivered by the LDO?

APPLICATION EXAMPLE - LDO



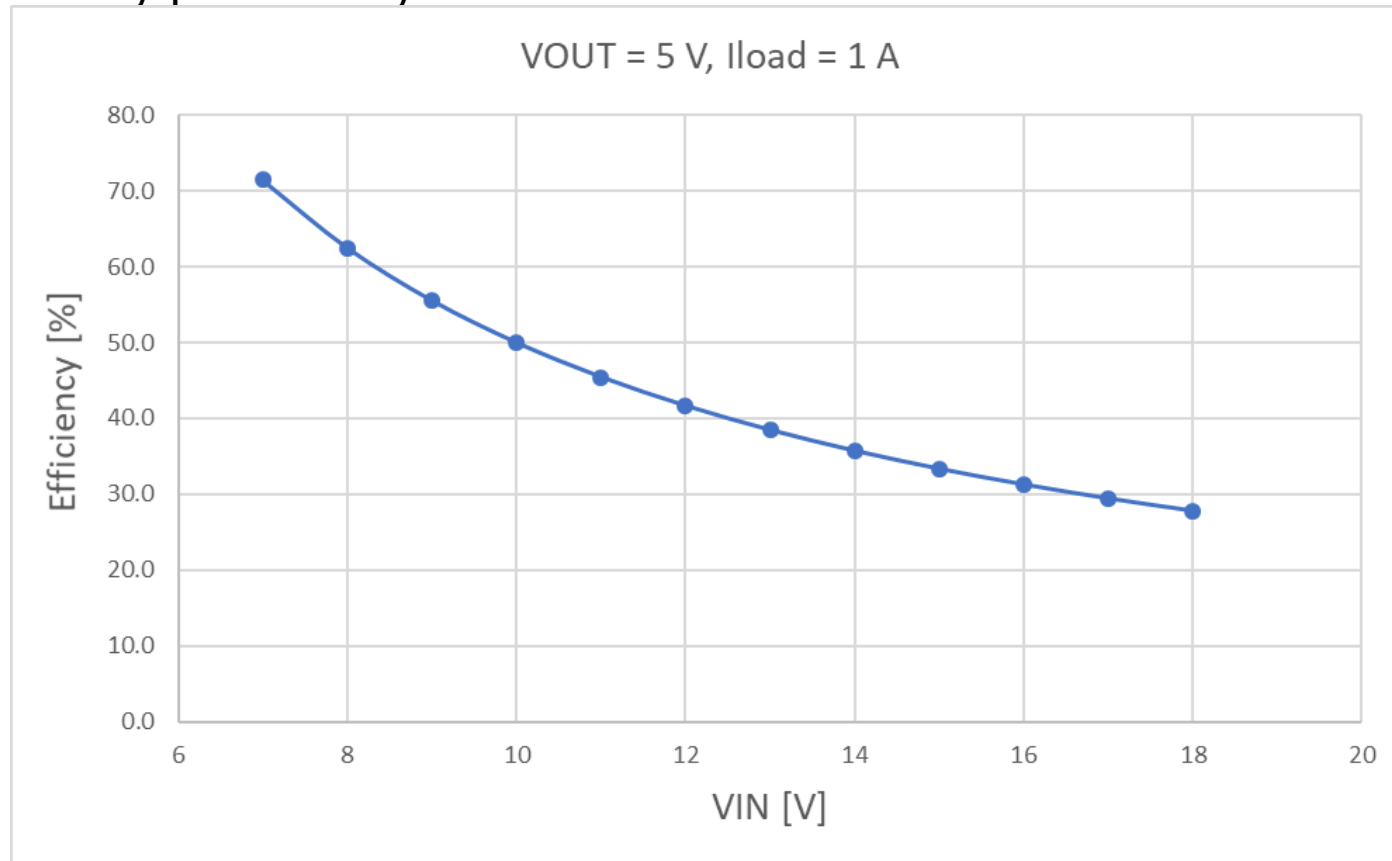
$$V_{OUT} = 200\text{mV} \cdot (1 + R2/R1) - (I_{ADJ} \cdot R2)$$
$$V_{ADJ} = 200\text{mV}$$
$$I_{ADJ} = 30\text{nA AT } 25^{\circ}\text{C}$$
$$\text{OUTPUT RANGE: } 0.2\text{V TO } 9.5\text{V}$$

LT3022 from Linear Technology

- Design the LDO supply circuit with LT3022 such that the LDO delivers 5 V
- Calculate and simulate the efficiency of the LDO for an input voltage of 10 V. Simulate the circuit in LTSpice.

DCDC CONVERTERS

LDOs offer a simple, low noise and cost-effective solution to supply active electronic components of an ECU. However, the efficiency becomes too low if the voltage difference between the output and the input is high. This leads to an increase of power loss and an increase of component temperature resulting in decreased component reliability and lifetime. Moreover, the efficiency of electronic systems is crucial for battery-powered systems



Alternative Solution with much better efficiency:

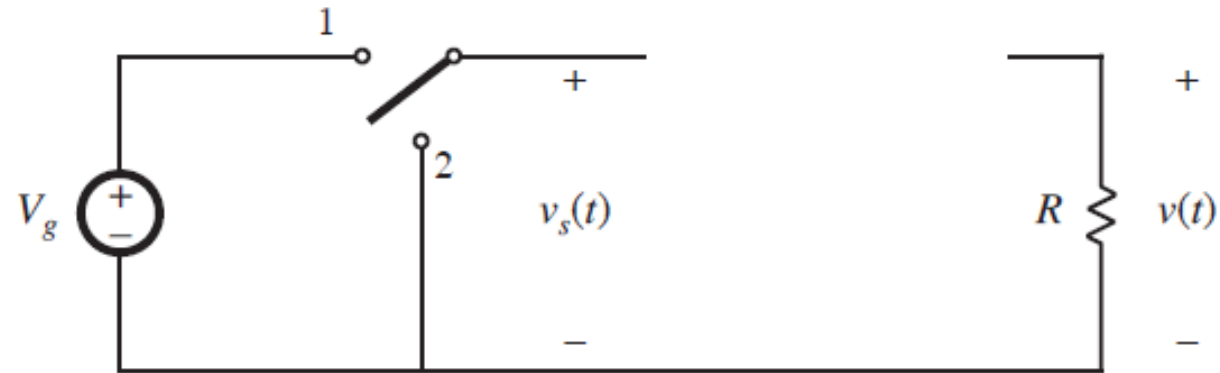
DCDC Converters



DCDC CONVERTERS

Buck Converter

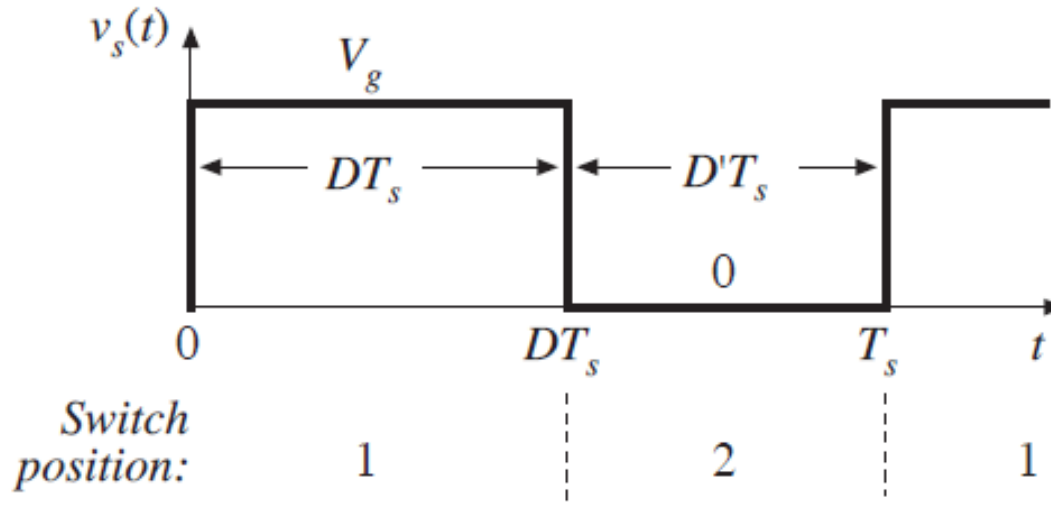
SPDT switch changes dc component



Switch output voltage waveform

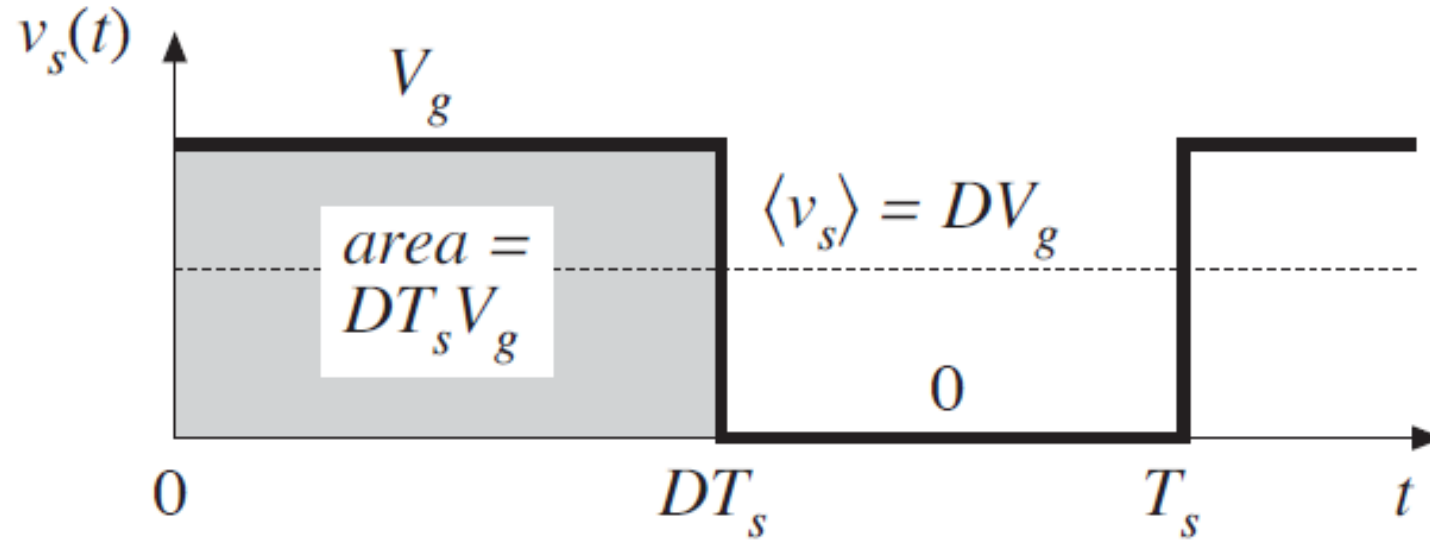
Duty cycle D :
 $0 \leq D \leq 1$

complement D' :
 $D' = 1 - D$



DCDC CONVERTERS

DC Component of a Buck Converter



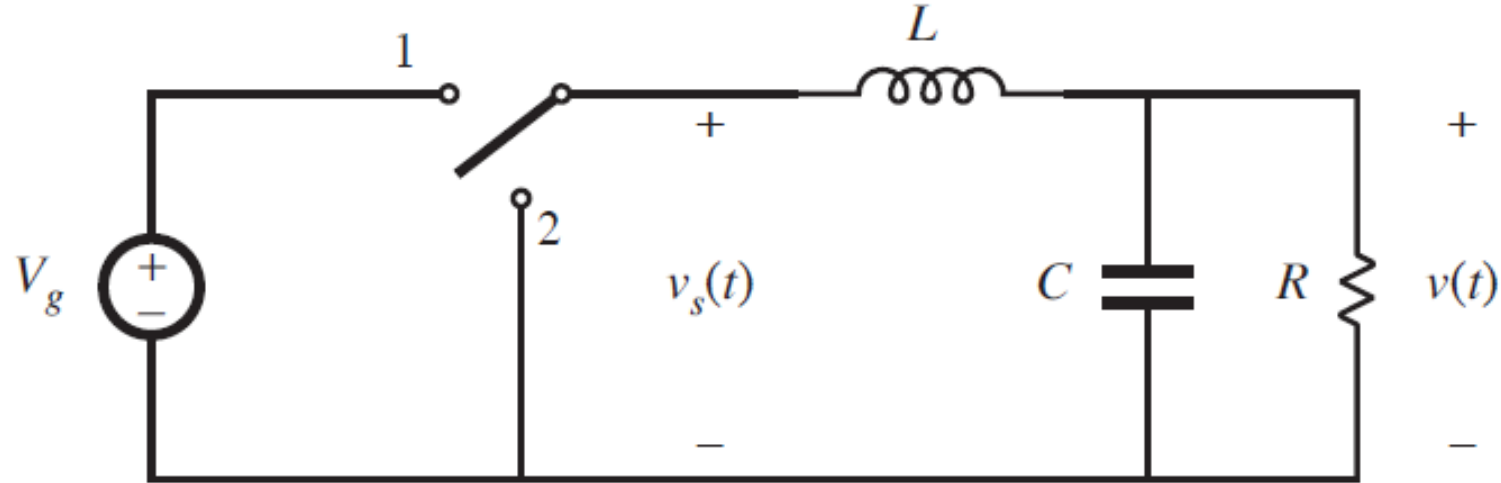
Fourier analysis: Dc component = average value

$$\langle v_s \rangle = \frac{1}{T_s} \int_0^{T_s} v_s(t) dt$$

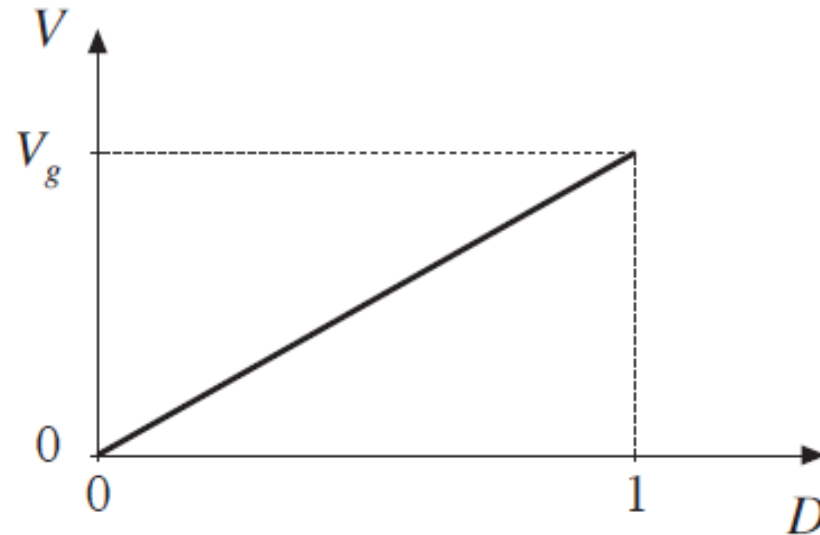
$$\langle v_s \rangle = \frac{1}{T_s} (DT_s V_g) = DV_g$$

DCDC CONVERTERS

Insertion of low-pass filter to remove switching harmonics and pass only dc component



$$v \approx \langle v_s \rangle = DV_g$$



DCDC CONVERTERS – ANALYSIS

Simple simulation testbench of a Buck Converter

$$\frac{V_{L_1}}{L_1} = \frac{di_{L_1}}{dt}$$

$$V_{L_1} = L_1 \frac{di_{L_1}}{dt}$$

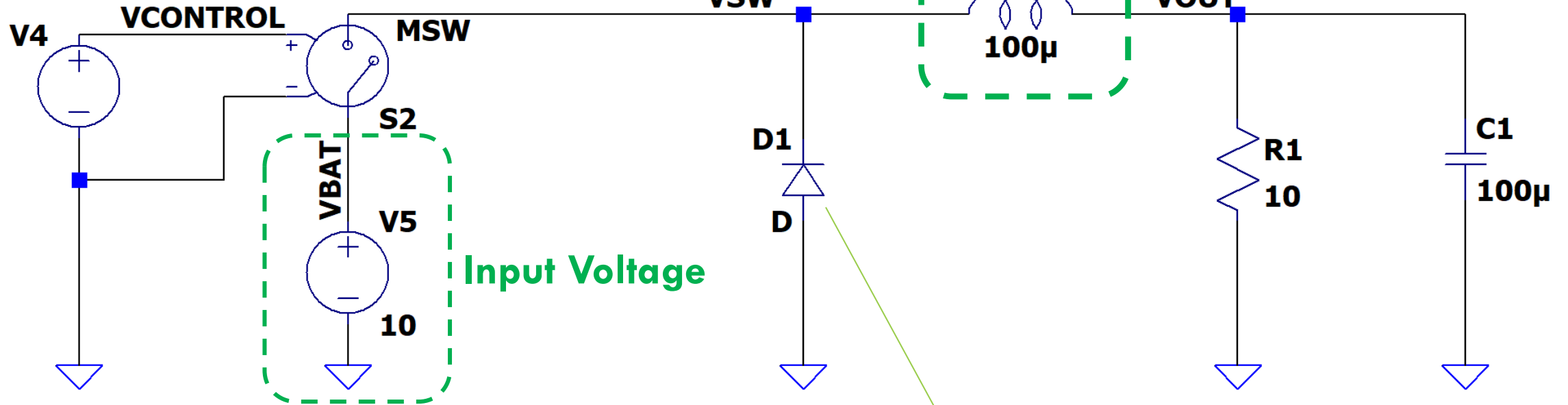
$$V_{L_1} = V_{SW} - V_{OUT}$$

$$i_{L_1} = \frac{1}{L_1} \int_0^{T/2} V_{L_1} dt$$

When the switch MSW (which is actually a transistor) is closed, calculate the current flowing through the inductor:

.tran 50m

PULSE(0 5 1n 100n 100n 0.005m 0.01m)

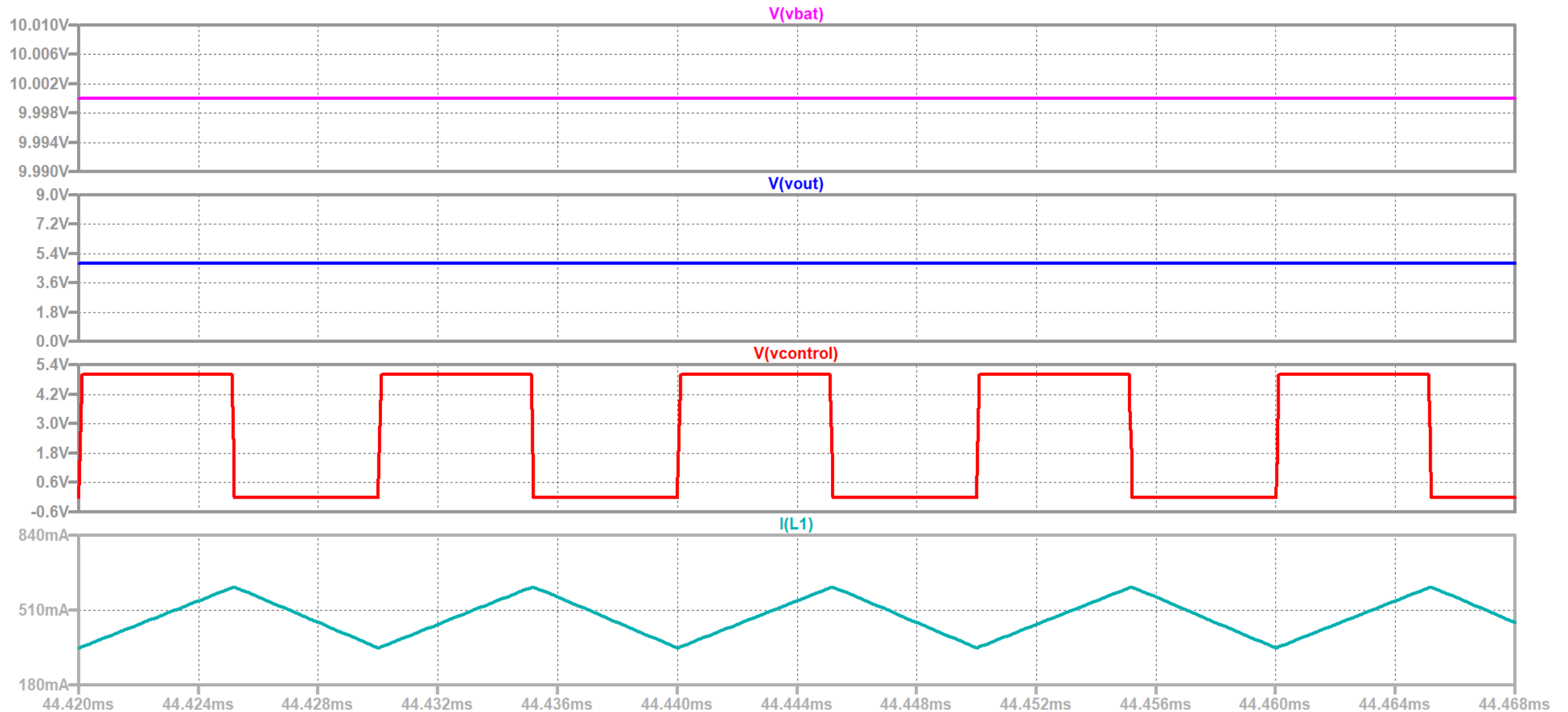


.model MSW SW(Ron=25u Roff=1G Vt=0.1 Vh=0.1)
.inc opamp.sub

D1 can be replaced with a MOSFET:
Complementary Switch

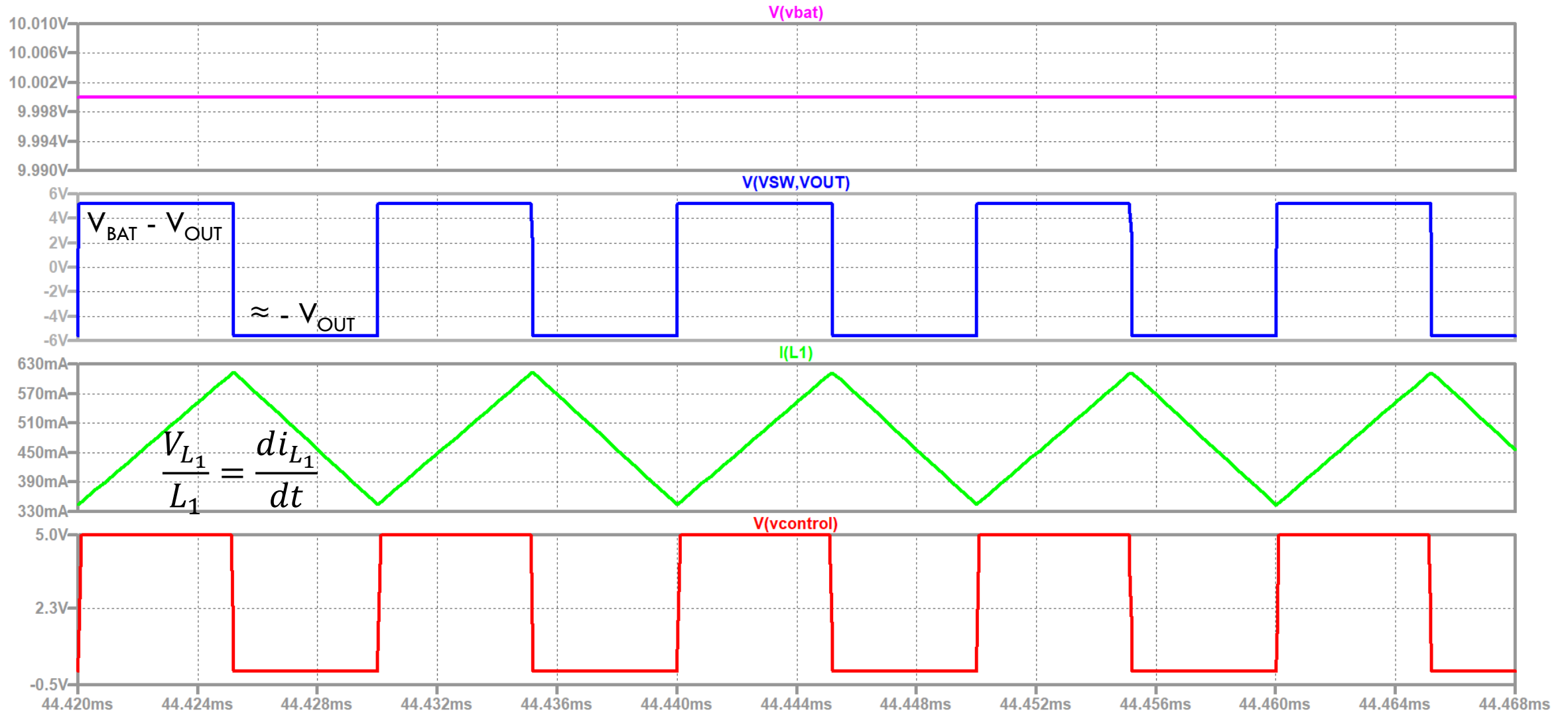
DCDC CONVERTERS – ANALYSIS

Simple simulation testbench of a Buck Converter

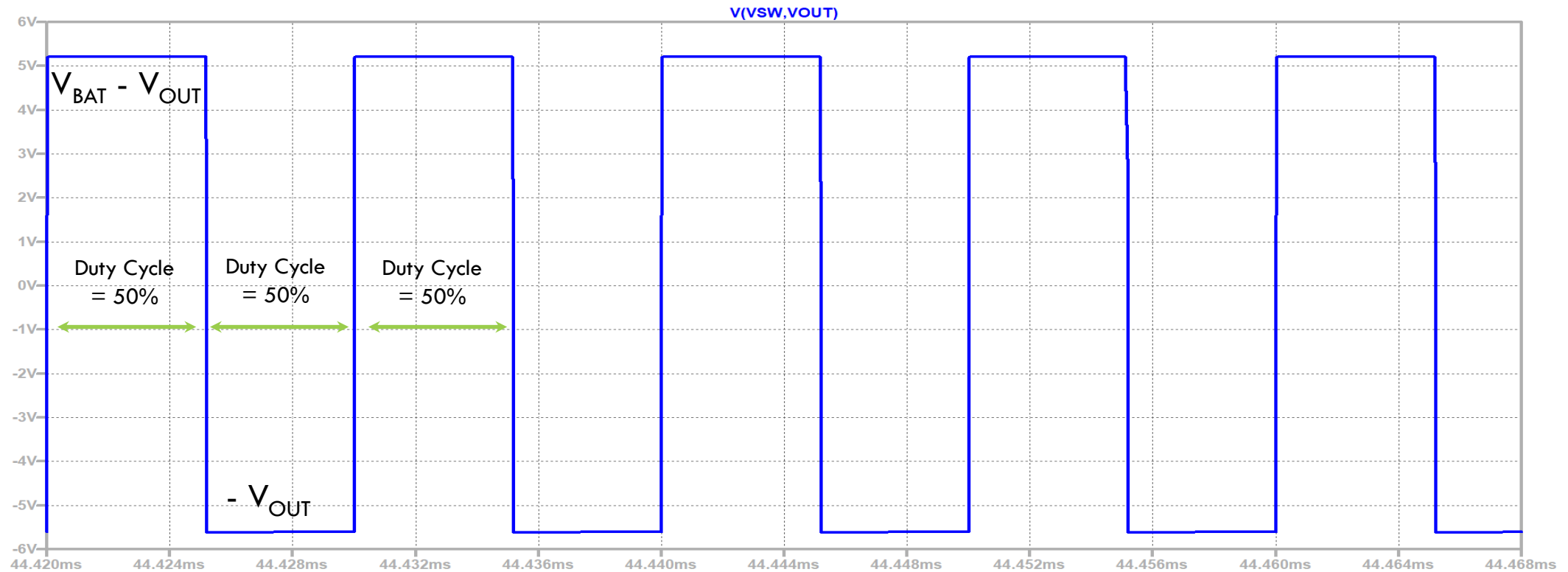


DCDC CONVERTERS – ANALYSIS

Simple simulation testbench of a Buck Converter



DCDC CONVERTERS – ANALYSIS



Average Voltage on the inductor is zero, if the duty cycle DC is 50%:

$$(V_{BAT} - V_{OUT}) \cdot DC - V_{OUT} \cdot (1 - DC) = 0$$

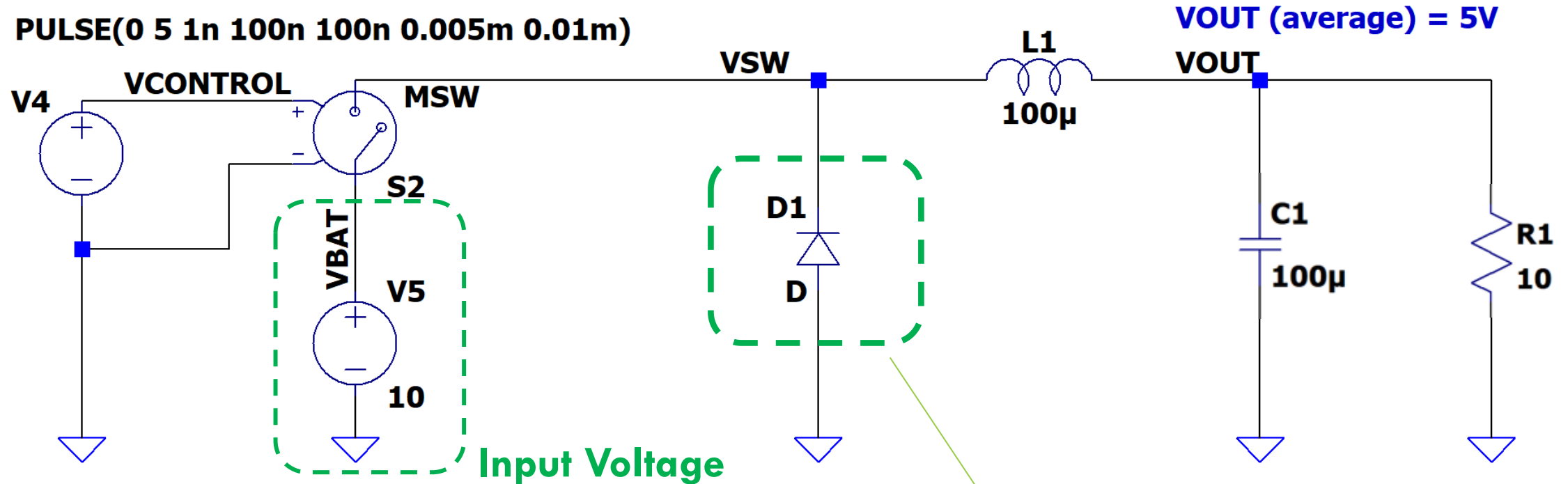
$$V_{OUT} = V_{BAT} \cdot DC$$

DCDC CONVERTERS – ANALYSIS

Simple simulation testbench of a Buck Converter

`.tran 50m`

`PULSE(0 5 1n 100n 100n 0.005m 0.01m)`

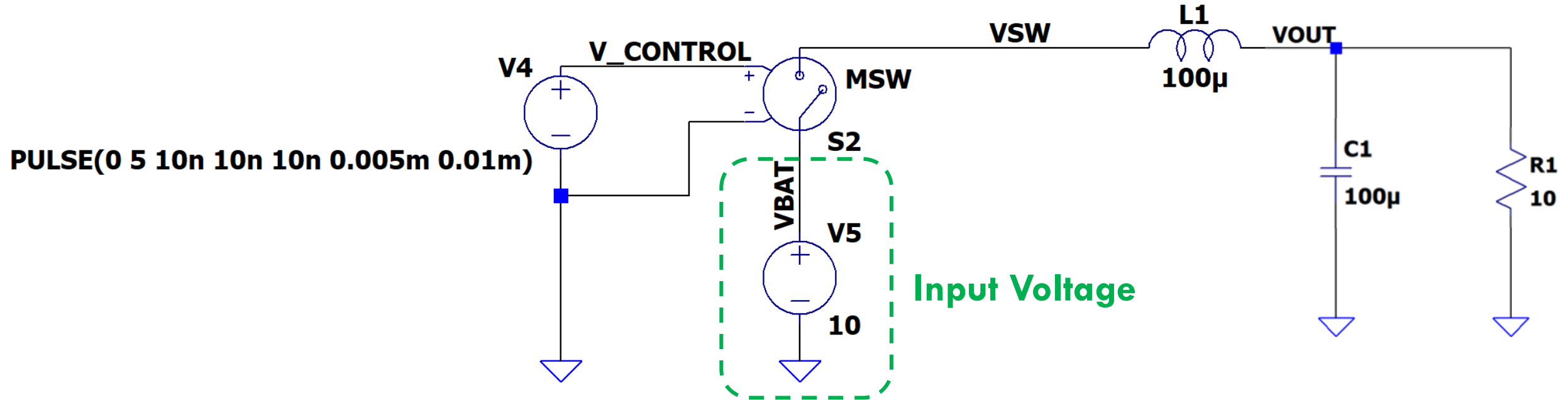


`.model MSW SW(Ron=25u Roff=1G Vt=0.1 Vh=0.1)`

`.inc opamp.sub`

Why do we need this diode?

DCDC CONVERTERS – ANALYSIS



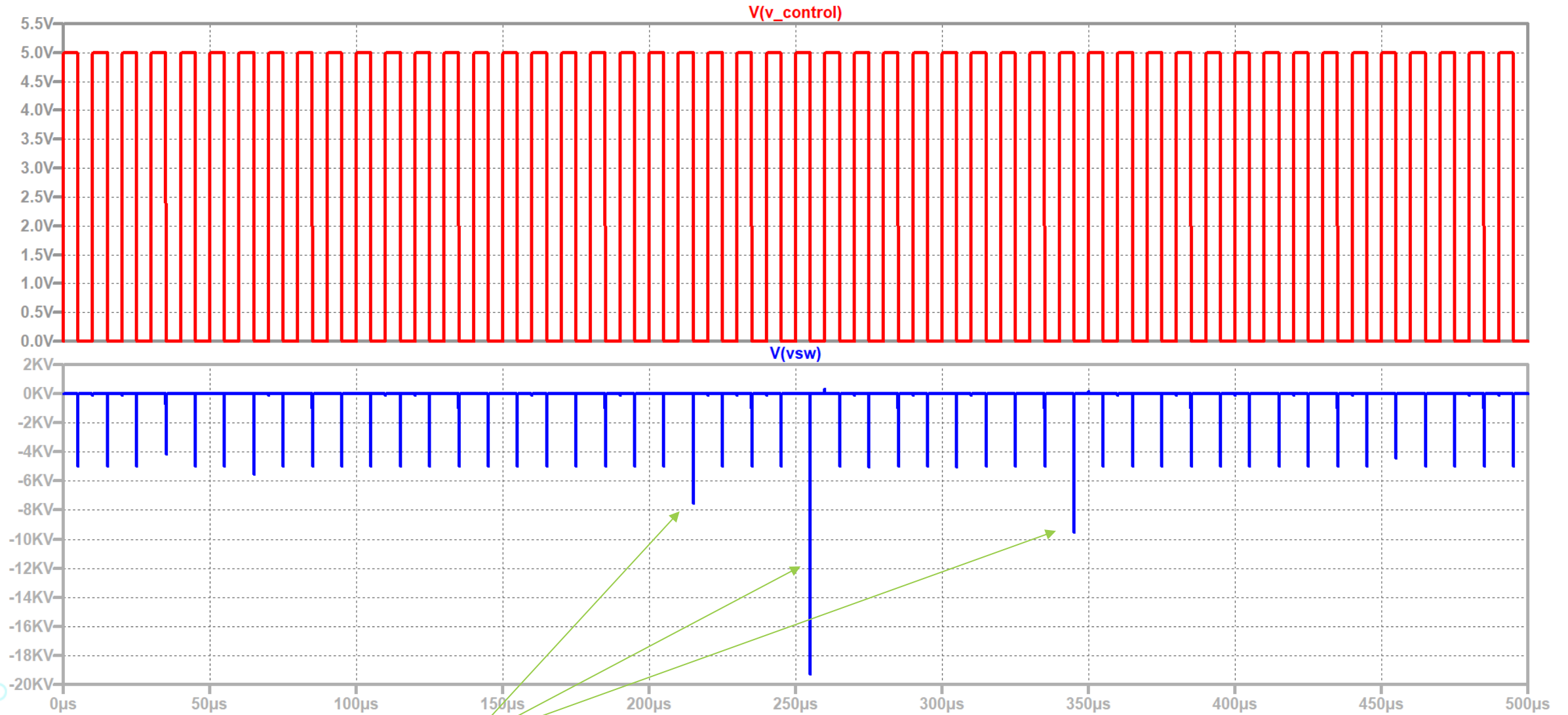
```
.model MSW SW(Ron=25u Roff=1G Vt=0.1 Vh=0.1)
```

```
.inc opamp.sub
```

```
.tran 0.5m
```

Diode removed! → what happens to VSW?

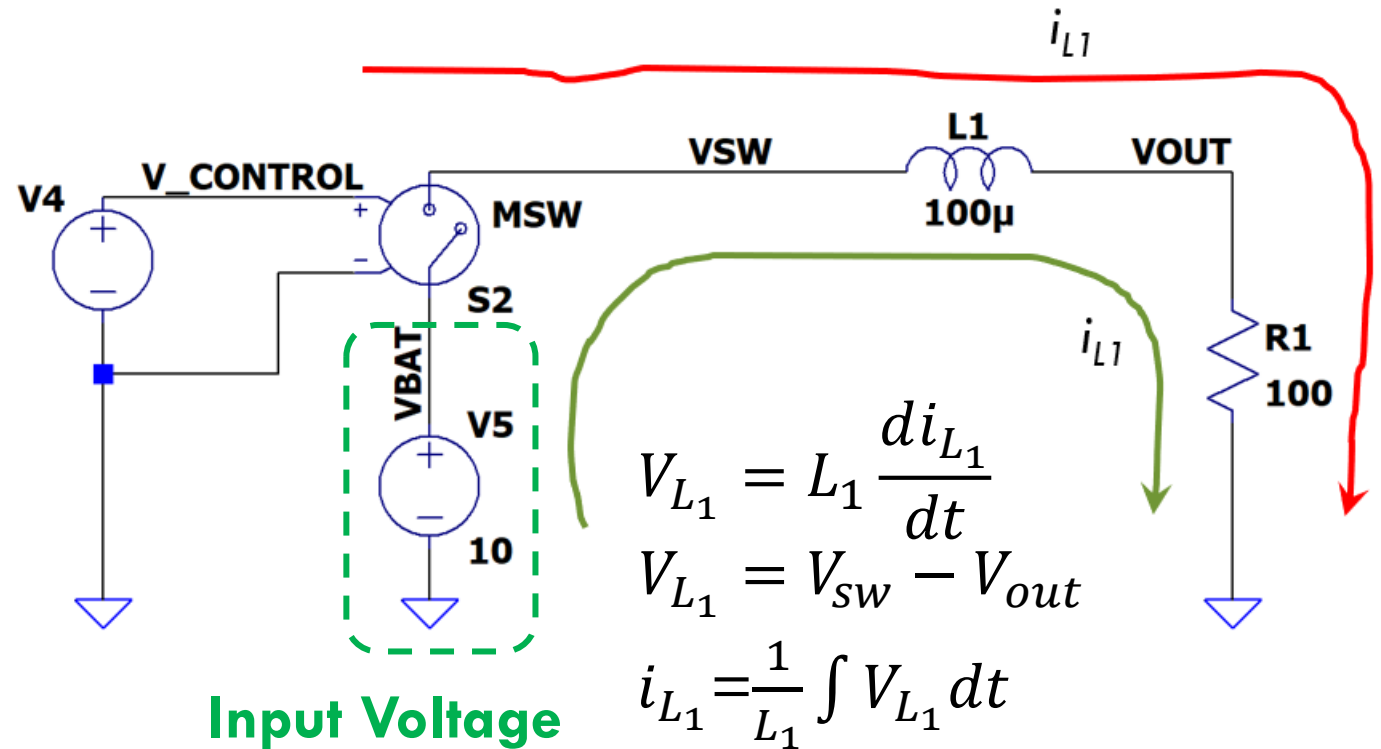
DCDC CONVERTERS – ANALYSIS



Extremely high negative voltages!!!

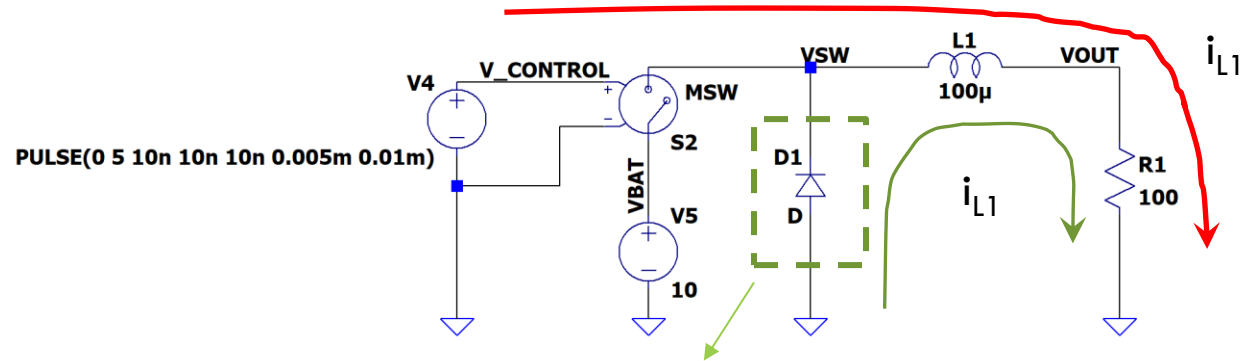
DCDC CONVERTERS – ANALYSIS

- An inductor will always try to keep the current constant!
- When the switch is turned off, the voltage source V4 is disconnected from the inductor. The inductor will try to keep the current i_{L1} constant and act as a voltage source
- The polarity of the inductor will change and VSW becomes negative and floating
- In a real-world application, this extremely high VSW voltage would lead damage the transistor, which is modeled by the switch (MSW)!



VSW becomes floating (not connected to a defined potential) when the switch is open!

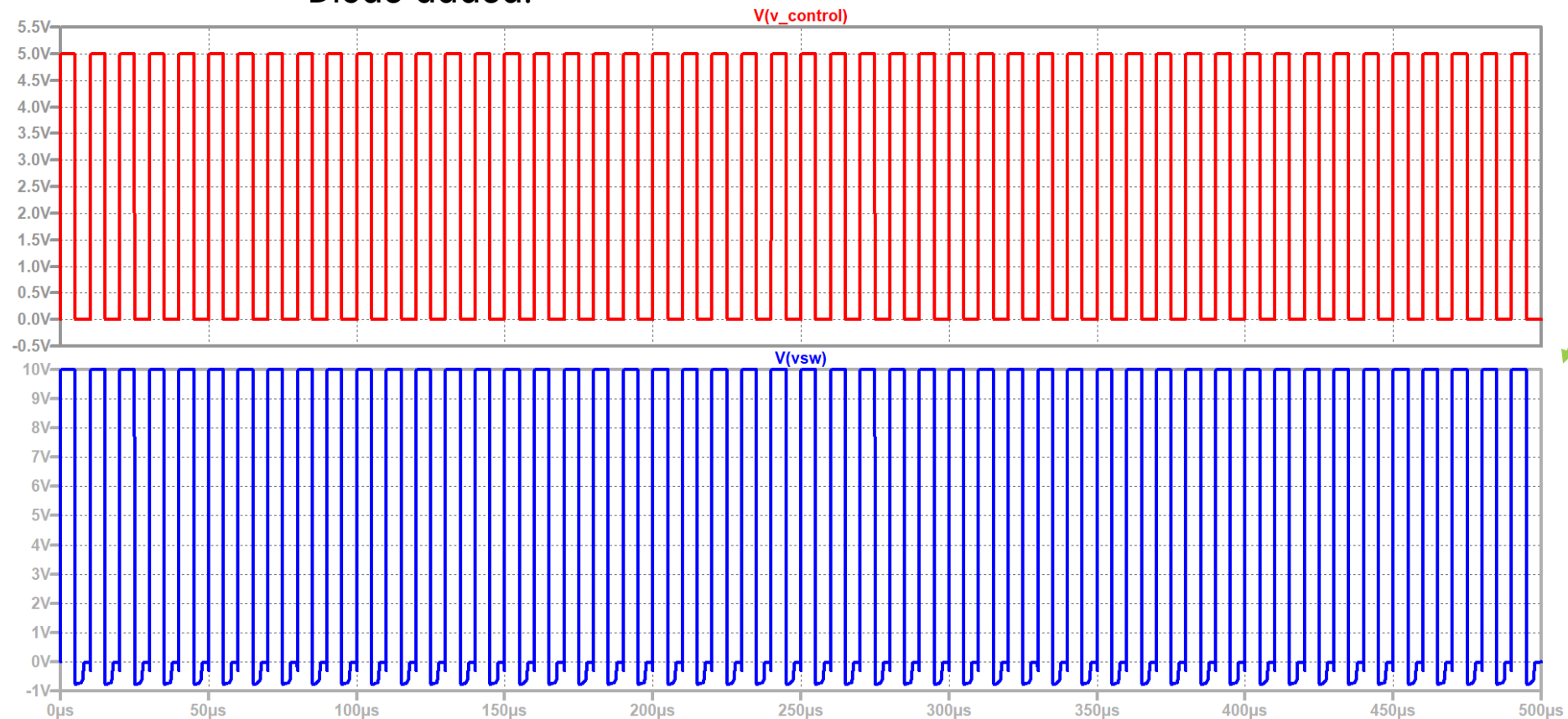
DCDC CONVERTERS – ANALYSIS



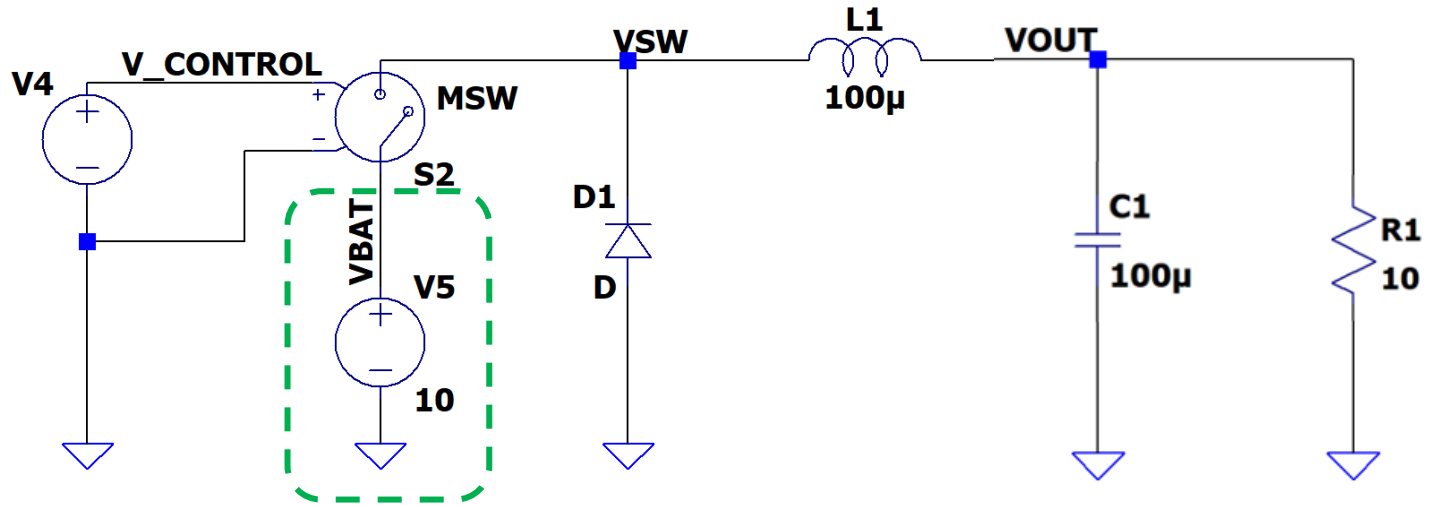
VSW is not floating anymore when the switch is open!

Diode added!

VSW peak voltage is 10 V!

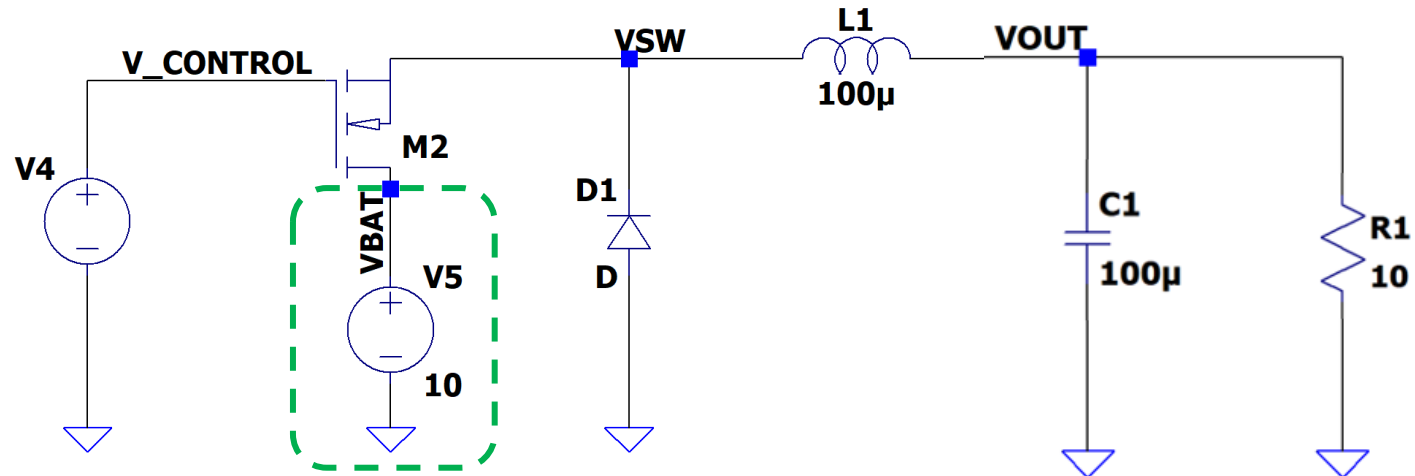


DCDC CONVERTERS – ANALYSIS



Input Voltage

Ideal switch replaced with a real MOSFET

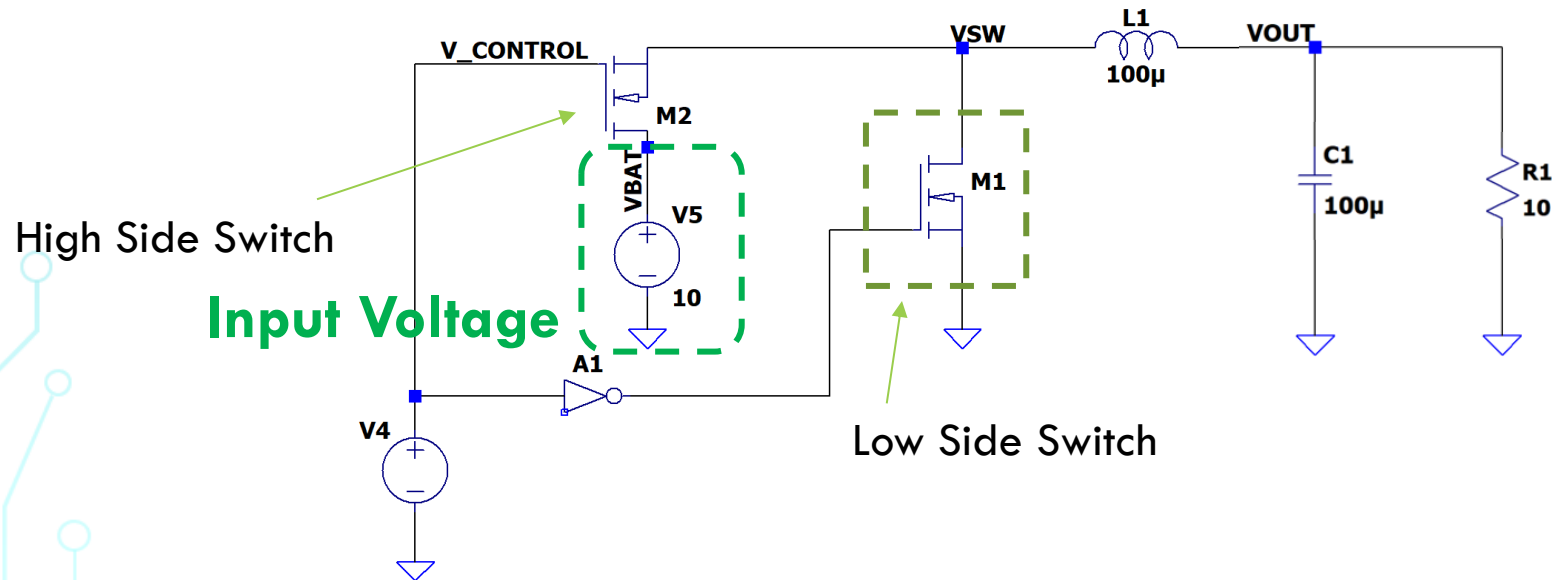
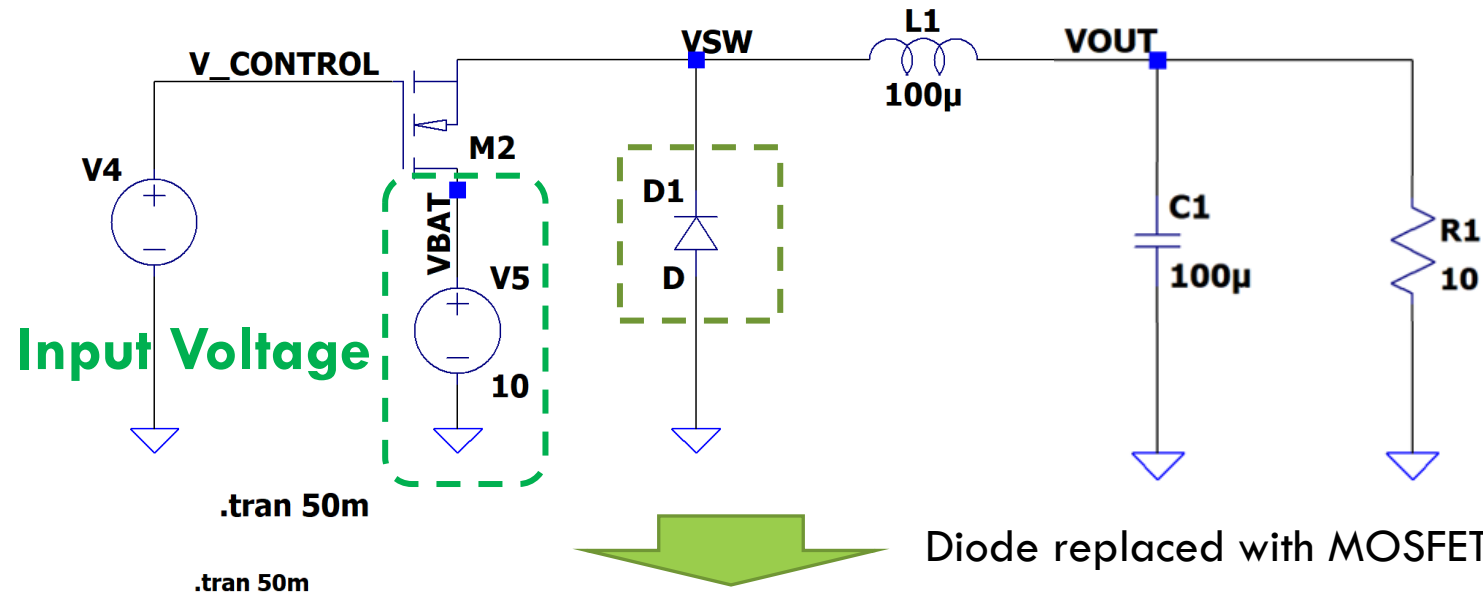


.tran 50m

Input Voltage

Diode cause a voltage drop of approximately 0.7 V. Is there a better alternative for the diode?

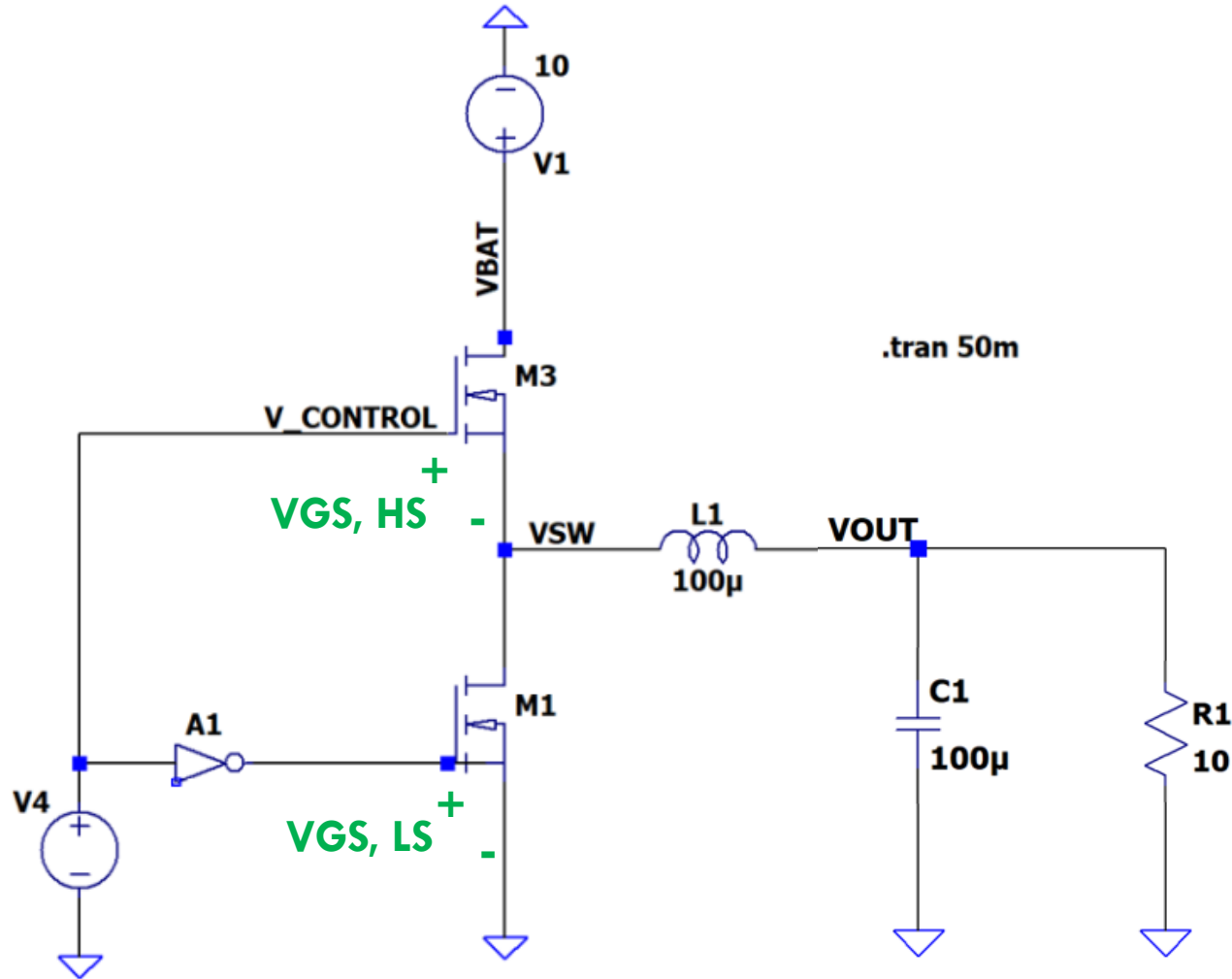
DCDC CONVERTERS – ANALYSIS



- If HS SW is on, then LS SW is off
- If HS SW is off, then LS SW is on

Complementary Switch

DCDC CONVERTERS – ANALYSIS

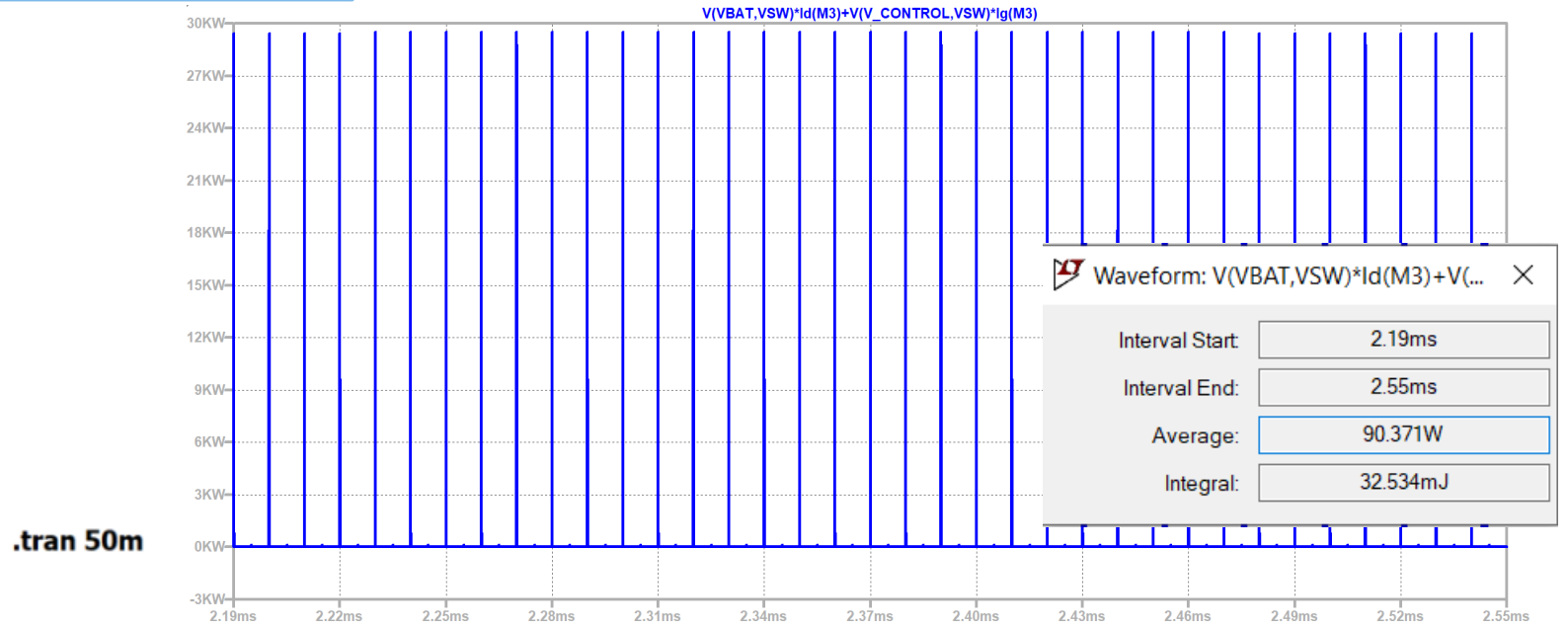
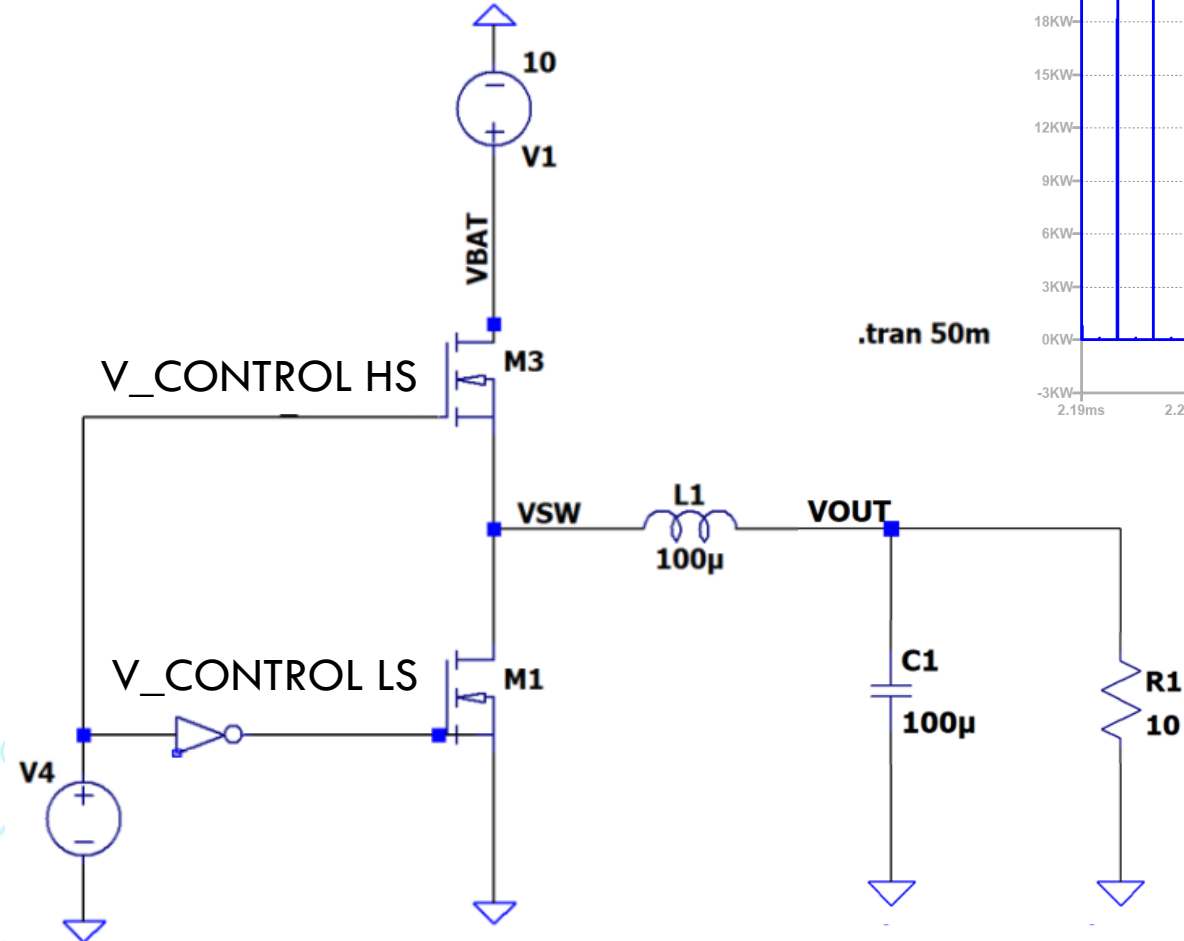


$$V_CONTROL \approx VGS_HS + VBAT$$

with $VGS_LS = VGS_HS$ if the MOSFETs used in HS and LS switches are the same

Complementary Half Bridge

DCDC CONVERTERS – ANALYSIS



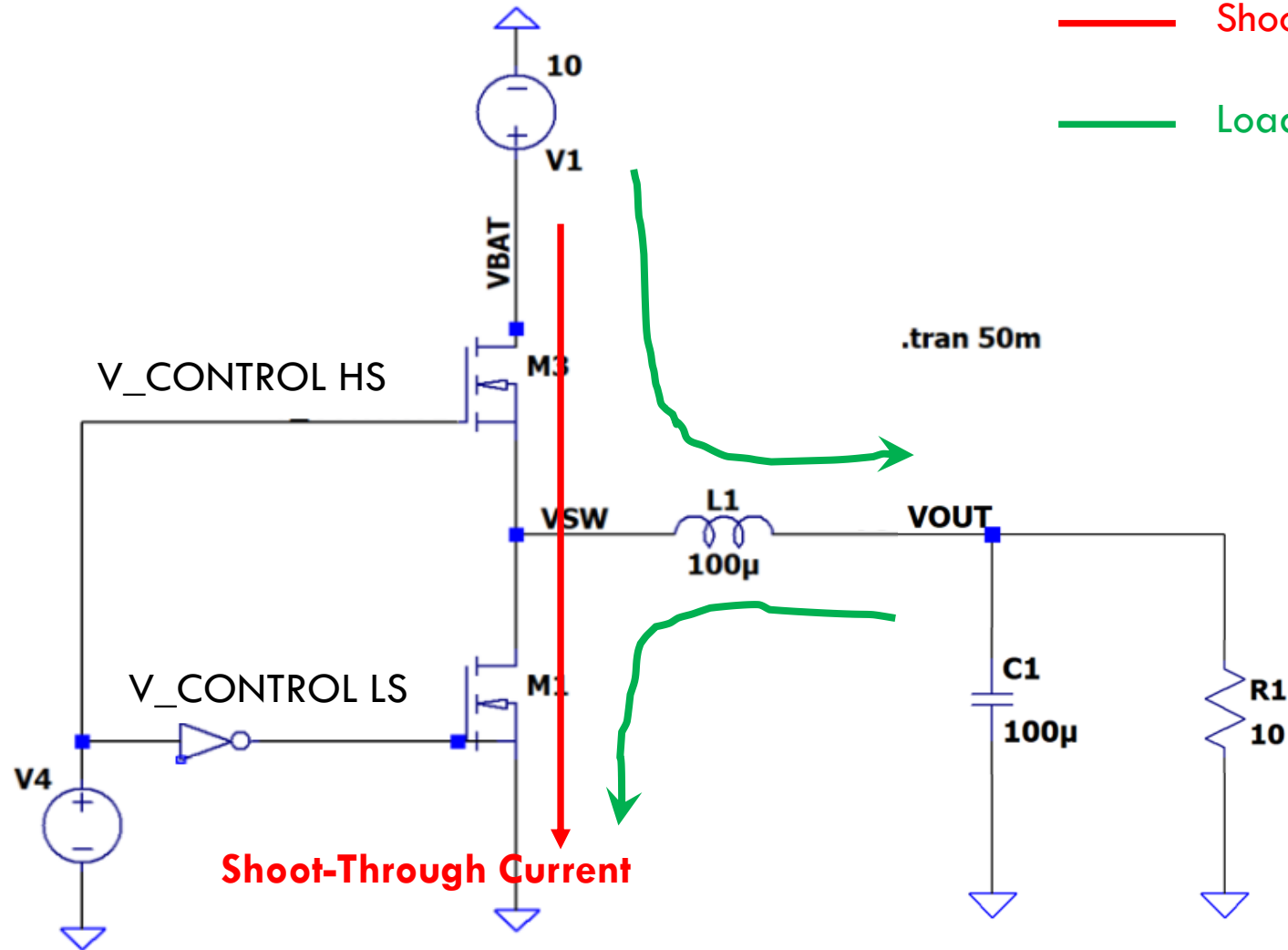
Power Dissipation of MOSFET = 90.37 W! $\rightarrow$ too high

DCDC CONVERTERS – ANALYSIS

HS and LS Switches conduct at the same time due to rise and fall times of the gate voltages → Shoot-Through!!!



DCDC CONVERTERS – ANALYSIS



— Shoot-through Current (undesired)

— Load Current (desired)

Shoot-Through currents occurs due to a short-circuit of (+) and (-) poles of the battery for a short time. This short-circuit leads to a steep increase of power dissipation of MOSFETs, which can damage the MOSFETs!!!

DCDC CONVERTERS – ANALYSIS

Shoot-Through Problem can be solved by introducing a so-called **Dead Time** in the PWM control signal



References

- Ribbens, William B. *Understanding automotive electronics* / William B. Ribbens; with contributions to previous editions by Norman P. Mansour ... [et al.] – 6th ed..
- Analysis and Design of Analog Integrated Circuits”, Gray Mayer et al, 5th Edition, Wiley, 2009
- Power Topologies Handbook, Markus Zehendner, Matthias Ulmann, Texas Instruments, 2020